

Wellmere

Season Lead · Wellmere Seed Bank

15 DAYS · 61 STOPS · GRADE 12 · SEEDBANK

THE OPENING CARD

Two hundred kilometres east there is a wheat disease the breeding programme has exactly one gene against. The wind that would carry it here comes in over the causeway. You are Season Lead at Wellmere. Forty-one thousand seed samples are held here at minus eighteen, and the ground outside is laid out in rings, spaced by how far pollen can travel. Sowing starts in three weeks, and what goes into the ground is settled this fortnight. The trial has to be laid out. The oldest samples have to be grown out again, before fewer of them will come up. And a new wheat has to be either released to the merchant or held back. Anything not sown in three weeks waits a full year.

What is actually in the vault

BEFORE THE DAY — GREET

Six leads and one season, and the crossing block is Thursday

Everybody here has one season a year and all of them want the same glasshouse slots. Volpe, the head breeder, wants you known to as many of them as you can before the plan is argued over, because a negotiation about bench space between strangers takes twice as long and lands worse.

PLAN CARD 113W

Monday in the first week of March, and the sowing plan has to be signed by Friday. Wellmere holds 41,000 accessions at minus 18, and Yusuf Mbaye's germination schedule says 60 of them have dropped below the line where they have to be grown out this season. There is room in the ground for about 40. Chiara Volpe wants that ground for her own trial and Nasrin Qureshi wants it for the collection. Today you find out what a seed is doing while it is stored, what the crossing programme is trying to make, and how the trial ground is laid out — because Friday's plan is made out of those three things.

BRIEFING 7W

41000 accessions, and 60 of them overdue.

WHAT YOU DECIDE 15W

Find out what Wellmere is holding and what it is doing while it sits there.

1.1 Why the vault is cold

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · BALLPARK

SCENE 34W

Mbaye has two lots of the same accession on the bench. 1 spent 4 years at minus 18 degrees and 6% moisture. The other sat in a warm shed and now germinates at 31%.

GUIDE 62W

Deterioration is a chemical rate, so cold and dryness each multiply the time a seed lasts, and the two multiply together. The shed lot is the baseline. Work out how much longer the vault lot should last, using the temperature factor and the moisture factor and nothing else — two of the tiles are quantities from a different part of the problem.

ASK 15W

How many times longer should the vault lot last than the lot in the shed?

BEHIND THE BUTTON 136W

Why a rate rather than a switch. A dry, cold seed is not dormant and not protected; it is still repairing damage and still accumulating it, only slowly. That is why storage life is measured in half-lives of viability and why a seed lot has no shelf date, only a rate.

What the two factors do separately. Around this range each 5 °C drop roughly doubles storage life, and each 2 percentage points of seed moisture removed does much the same. Neither kills anything and neither stops anything — both slow the same reactions.

Why they multiply. The reactions are the same reactions, and each factor divides their rate, so the effects compound rather than add. That is what makes a cold dry vault worth its electricity: sixty-four times is a different institution from eight times.

TAKES AS READ

orthodox seeds are seeds that can survive drying and cold storage · seed viability as a rate: what a stored seed is still doing — taken as read

VERDICT 128W

Cold, dry storage works on a rate. A stored seed is still repairing damage and still taking it on. Both variables slow the same reactions. The shed sat at about 2 °C, with seed at 14 per cent moisture. The vault holds minus eighteen and 6 per cent. That is 20 °C colder. At roughly a doubling per 5 °C, it is 2⁴, or 16 times. It is also 8 points drier. Halve the moisture again and the same thing happens, so that is another few doublings. Both factors divide the rate of one process, so they multiply rather than add. The answer is about 64 times, not 32. That gap is the difference between holding an accession for a working lifetime and holding it for a career.

TAKEAWAY 14W

Cold and low moisture extend orthodox-seed life by slowing the reactions that drive deterioration.

1.2 Two parents, and what you are asking of them

HEREDITY & CROSSING · CROSSING HALL · A PERSON · CHOICE

SCENE 36W

Volpe has two parents pinned above the bench. One is a high-yielding modern variety. The other is a low-yielding landrace that survived the 2018 drought. Otto Bengtsson asks what combination the programme is trying to create.

GUIDE 0W

ASK 11W

What is the programme actually asking for by crossing these two?

TAKES AS READ

offspring get one set of chromosomes from each parent

VERDICT 79W

Meiosis gives each gamete one copy of each chromosome. The F1 therefore receives one set from each parent. The next generation reshuffles those copies into many combinations. The breeder wants offspring carrying useful yield alleles from the modern parent and drought alleles from the landrace. Neither parent has that combination. Many offspring will be worse than either parent because useful and harmful alleles also segregate. Breeding works by making enough recombinant offspring to find the rare combinations worth keeping.

TAKEAWAY 19W

A cross is a request for a combination, and it takes generations to find out whether it was granted.

1.3 Why the same line is sown four times

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · CHOICE

SCENE 37W

Meera Quiroga has the plot map on the wall: 240 plots, with each line appearing 4 times across 4 blocks. The order is randomised, and the north end of the field is visibly wetter than the south.

GUIDE 0W

ASK 10W

What does sowing each line in four randomised plots buy?

TAKES AS READ

the same crop grows differently in different parts of a field

VERDICT 81W

A field varies in drainage, soil depth, compaction and shelter. Sow a line once and its yield contains both genotype and the conditions of that plot. Replicating the line in several blocks spreads it across different ground. Randomising positions prevents the researcher's habits from deciding which line receives favourable places.

Differences among replicate plots estimate environmental variation. That variation becomes part of the uncertainty when lines are compared. Replication therefore improves the genetic comparison rather than the measurement of one plot.

TAKEAWAY 19W

A field is not uniform, so a single plot measures the plot as much as the line in it.

END OF THE DAY 19W

A stored seed is alive and spending itself, and everything in the vault is aimed at slowing that down.

What the generation tells you

BEFORE THE DAY — TRIAL

Walk the inner rings before the sowing plan is signed

The station is laid out by isolation distance — vault at the middle, then the lab, then the glasshouses — and the plan you are about to sign puts material in specific rings. Qureshi, the collection curator, walks every new hand round the inner ones once, because a sowing plan drawn from a map ignores which way the wind crosses the headland.

PLAN CARD 99W

Tuesday. The F2 of Volpe's 2024 cross is in the warm bay — 400 plants, and the first generation of any cross where the parents' alleles come apart and show themselves. Hana Nakagawa has run a marker on the same plants and has a disagreement to explain: 11 of them score as carrying the resistance allele and do not look it. Osman Toft, meanwhile, wants somebody to explain to the trainees where the grain in his drying hall came from. Today you count what the generation is telling you, decide what the marker mismatch means, and follow the carbon.

BRIEFING 11W

400 plants in the warm bay, and a ratio to count.

WHAT YOU DECIDE 14W

Read the second generation of last year's cross, and check it against a marker.

2.1 Counting a generation

HEREDITY & CROSSING · CROSSING HALL · A ROOM · BALLPARK

SCENE 30W

400 F₂ plants, and 306 of them show the resistant reaction while 94 do not. Bengtsson asks what that number is supposed to be, and whether it is close enough.

GUIDE 66W

Work out what a single dominant locus predicts, then compare. Two heterozygous parents give three resistant to one susceptible, so the expected count is the number of plants times three quarters. The two-gene alternative, where either dominant allele is enough, predicts fifteen to one instead. Those two expectations are far apart at four hundred plants, which is what makes the observed count evidence rather than decoration.

ASK 19W

How many of the 400 F₂ plants should show the resistant reaction if a single dominant locus governs it?

BEHIND THE BUTTON 166W

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a quantity belonging to a different part of the problem is caught before it is placed — and two of the tiles here belong to the alternative model, not to this one.

Where the three quarters comes from. Each parent carries one resistant and one susceptible allele, and meiosis puts one of the two into each gamete. Four equally likely combinations follow, and three of them contain at least one resistant allele. Dominance is what makes those three look alike.

Why the comparison needs both models. A count near three hundred is only evidence for one locus because a rival model predicts something else. Fifteen to one puts twenty five susceptible plants in four hundred rather than a hundred, so a single tray of this size separates them. A ratio quoted without the alternative it beats is a number nobody can argue with.

TAKES AS READ

a plant carries two copies of each gene and passes on one

VERDICT 105W

F₁ plants carry one resistance allele and one susceptible allele. Meiosis separates those alleles, so 2 heterozygous parents give a 3 : 1 phenotype expectation when resistance is dominant. In 400 offspring that is $400 \times 3 \div 4 = 300$ resistant and 100 susceptible. The observed 306 and 94 sit close to it. 2 independently assorting genes where either dominant allele gives resistance predict 15 : 1, which is $400 \times 15 \div 16 = 375$ resistant and 25 susceptible. The counts therefore support one dominant locus and rule the two-locus alternative out, which no single plant in the tray could have told you.

TAKEAWAY 15W

The proportions in a whole generation carry information that no single plant in it does.

2.2 One disagreement, two possible causes

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A PERSON · DEGENERACY

SCENE 40W

Nakagawa has 400 F2 plants scored by a linked PCR marker and by resistance. 11 carry the marker but fail the resistance test. The PCR controls and gel bands look clean; the marker lies about four centimorgans from the gene.

GUIDE 53W

Two controls slide recombination rate and test error along a family of combinations that all produce eleven disagreements in four hundred plants. See how wide that family is before naming a cause. Then bring in the direct-gene test, which fails for one of those explanations and not the other, and watch it collapse.

ASK 16W

Which combinations can explain the first disagreement rate, and which one survives the direct-gene PCR check?

BEHIND THE BUTTON 142W

Why one disagreement rate cannot separate the two. A marker four centimorgans from the gene is expected to be separated from it in about four plants per hundred by recombination. A test that occasionally misreads produces the same disagreements. Eleven in four hundred is consistent with several mixtures of the two.

What the direct-gene test does. It asks about the gene itself rather than about the marker, so recombination between them is irrelevant to it. Plants that carry the gene and failed the resistance test are test error; plants that lack the gene are recombinants.

Why the clean controls are not an answer. Controls show the reaction worked on known material. They cannot detect a plant-specific failure — poor template, an inhibitor, a mislabelled well — which is exactly the kind of error that produces a handful of disagreements in four hundred.

TAKES AS READ

PCR uses primers to amplify a chosen DNA region · gel electrophoresis separates DNA fragments by size · a nearby marker can be separated from a gene by crossing over · independent assortment, and the ratios two genes give instead — taken as read

VERDICT 77W

PCR copies a chosen DNA region through repeated denaturation, primer annealing, and extension. Gel electrophoresis can then separate the resulting DNA fragments by size. Here that tool reads a marker near the resistance gene, not the gene itself. Marker–trait disagreement can therefore come from recombination or assay error. One disagreement rate cannot separate those causes. A second PCR assay aimed

inside the resistance gene adds independent evidence. When it confirms the gene call, linkage and recombination remain.

TAKEAWAY 19W

PCR can identify a DNA sequence, but a linked DNA marker can still separate from the gene it predicts.

2.3 Where a tonne of grain comes from

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM · BALANCE

SCENE 36W

Toft has 11 tonnes from 1 field on the weighbridge slip and a trainee who has just said the crop got it from the soil. He would like that answered properly before it goes any further.

GUIDE 62W

Close the ledger. The rows are what the field took up or gave off over the season, and the total is the eleven tonnes of dry matter on the slip. Count the ones that are actually part of that dry weight and leave out the one that is a different quantity altogether. Reading a row costs nothing; counting it is the claim.

ASK 23W

Close the season's dry-matter ledger for the field. Which flows make up the 11 tonnes, and what is the term nobody wrote down?

BEHIND THE BUTTON 151W

What dry matter is. Grain is weighed at fifteen per cent moisture and the ledger is kept dry, so the water in the standing crop is not in the total at all. About nine tenths of what remains is carbohydrate built from carbon dioxide.

Why one row cannot be counted. The season's transpiration is four thousand tonnes of water, and it is not a component of dry mass — it is the flow that carried the minerals up and kept the leaves cool. Adding it to a mass ledger gives a number two orders of magnitude wrong and no warning.

What the minerals do. Nitrogen, phosphorus and potassium decide how much tissue the crop can build and contribute a few per cent of its weight. That is the distinction the trainee is missing: what controls the size of the harvest is not the same as what the harvest is made of.

TAKES AS READ

plants build sugars from carbon dioxide and water using light

VERDICT 143W

The eleven tonnes on the slip came out of the air. Photosynthesis fixed about 19.6 tonnes of dry matter over the season, respiration burned 6.1 of it back to carbon dioxide, and minerals from the soil account for 0.3 — a few per cent, which is why the trainee's answer sounds plausible and is wrong by an order of magnitude. The ledger still does not close, because 2.4 tonnes of it is root and stubble that stayed in the field: harvest index is the fraction of what the crop built that leaves in the trailer, and it is never one. The transpired water is the row that cannot be counted at all. Four thousand tonnes of it moved through the crop, and none of it is dry mass — it is the flow that carried the minerals and kept the leaves from cooking.

TAKEAWAY 16W

A harvest weighs what it weighs because of something that was not there when it started.

END OF THE DAY 16W

A ratio is evidence about what the parents carried, not a decoration on a data sheet.

What a collection is made of

BEFORE THE DAY — FOLLOW

Quiroga walks the trial rows once, before scoring starts

The trial is four blocks of twenty lines and the plot map is only true if you know where the blocks start. Quiroga, the field trials manager, walks it once at the pace scoring actually happens. Stay with her — she stops at the row ends, which is where the map and the ground disagree.

PLAN CARD 90W

Wednesday. Tomas Ivanescu has genotyped a sample from 1 landrace accession — the 1 Idris Fenn's grandfather gave Wellmere in 1994 — and has put the numbers on the board rather than a description. Sylvie Pirelli has found that two accession numbers point at the same collection site, which may mean one of them is a duplicate taking a regeneration slot. Today you learn what a population is made of, what the numbers would look like if nothing were acting on it, and how many plants a grow-out actually needs.

BRIEFING 3W

Not varieties. Frequencies.

WHAT YOU DECIDE 11W

Count what the vault holds at the level that actually matters.

3.1 Frequencies, not plants

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A ROOM · BALLPARK

SCENE 36W

Ivanescu has 100 plants from the 1994 accession scored at 1 locus. 36 are AA, 48 are Aa and 16 are aa. He asks for the allele frequency, then points to Hardy–Weinberg as the comparison model.

GUIDE 54W

Five numbers, and two of them count plants rather than allele copies: the sample size, and the recessive homozygotes. Ask of each whether this frequency needs it. A hundred diploid plants carry two hundred copies, and a heterozygote contributes one of them. Genotype counts and allele counts are different arithmetic on the same plants.

ASK 10W

Estimate the frequency of the tall allele in this accession.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a diploid plant carries two copies of the locus · Hardy–Weinberg is a null model that predicts genotype frequencies from allele frequencies

VERDICT 72W

100 diploid plants carry 200 allele copies. The 36 AA plants contribute 72 A copies, and the 48 heterozygotes contribute 48 more. That gives $p = 120/200 = 0.60$ and $q = 0.40$. Hardy–Weinberg predicts $p^2 = 0.36$, $2pq = 0.48$ and $q^2 = 0.16$. Those values exactly match the observed genotype fractions here. This gives a clean null model before drift, selection, migration, non-random mating or a bottleneck changes the population.

TAKEAWAY 15W

A frequency describes the population; a genotype describes one plant, and they answer different questions.

3.2 How many plants make a population

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · CHOICE

SCENE 30W

Mbaye can regenerate the 1994 accession from whatever number of plants the ground allows. Volpe points out that 30 plants would free half a plot. Ivanescu has gone very quiet.

GUIDE 54W

Four things a small grow-out might cost. Ask of each whether careful storage afterwards could repair it. Rare variants are the easiest to miss, because few plants carry them. Once an allele is in none of the contributors, no freezer restores it. And the resulting seed germinates well, which is why nothing routine notices.

ASK 10W

What does regenerating from 30 plants instead of 200 cost?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

each generation's alleles are a sample of the last generation's · genetic drift, and why it is worst in a small sample — taken as read

VERDICT 85W

Every regeneration samples alleles from the population that came before it. Rare variants are easiest to miss because few plants carry them. If an observable type occurs in only a few plants, a small grow-out can contain none of those plants by chance. Once the allele is absent from every contributor, careful storage cannot restore it. The resulting seed may still germinate well and look normal. That is why a genetic bottleneck can damage a collection without leaving an obvious sign in routine viability tests.

TAKEAWAY 18W

A grow-out is a sample, and a small sample loses whatever was rare in what it came from.

3.3 Move the line, not the story

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · CONTROL

SCENE 38W

Quiroga has three microplot positions across the slope and enough seed from two numbered lines for a quick repeat. The trainee's strip layout put each line in only one place. Yield scatter from repeated microplots is already known.

GUIDE 68W

The number you are watching is plot yield, and repeated microplots already tell you how much it scatters on its own. Three things about the setup can be changed: which line is sown, where on the slope it sits, and how thickly. A change counts only if it moves yield by more than that scatter. Change one, run the microplot, restore it, and name the variable yield follows.

ASK 17W

Which variable moves yield by more than the repeat scatter and then returns when you restore it?

BEHIND THE BUTTON 166W

Why the trainee's layout cannot answer the question. Each line was sown in one place, so line and position changed together. If the numbers differ, nothing in the data can say whether the better line or the better ground did it. That is what a confound is, and it is not fixed by more careful measurement of the same plots.

What the repeat scatter is for. Sow the same line twice in the same conditions and the two plots do not match: soil varies, weather varies, a hare gets in. That spread is the yardstick. A difference smaller than it is a measurement of the field rather than of the treatment.

Why position is the one to suspect on a slope. Water, depth and warmth all change down a slope, and all three move yield. A field trial is laid out to break the link between treatment and position on purpose — blocks, randomisation, replication — and this stop is what happens when that was skipped.

TAKES AS READ

field position can change crop yield · a causal comparison changes one candidate cause while holding the others fixed · dominance, and why one allele is enough — taken as read

VERDICT 81W

A strip design changes genotype and ground together, so either can explain the yield difference. The control run holds the line and sowing density fixed while moving only the plot position. That change shifts yield far beyond the repeat scatter. Restoring the original position restores the reading. Changing the line on the same ground produces only a small shift in this authored test. The result shows why blocks and randomisation are needed before genotype can be credited for a field difference.

TAKEAWAY 23W

When genotype and field position always change together, their effects are confounded; a control test separates them by changing one at a time.

3.4 Six per cent, while the lorries arrive

CONSERVATION & VIABILITY · SEED VAULT · A PERSON · HOLD

SCENE 39W

The drying room has to bring this week's intake to six per cent moisture before any of it goes into the vault. Wet lots keep arriving off the field, and the dehumidifier is the only thing that answers them.

GUIDE 49W

Hold the room's moisture inside the band drawn on the dial. The band narrows as the week goes on, because seed that is nearly dry is spoiled by less. The dehumidifier is your control, and each lot that arrives is a load that keeps pushing until something answers it.

ASK 13W

Hold the drying room at six per cent while the week's intake arrives.

BEHIND THE BUTTON 146W

Why the band narrows. Early in the week the intake is wet and a point either way changes nothing. By the end the lots are close to store moisture, and the same point is the difference between seed that keeps for decades and seed that keeps for years.

Why a wet lot is not a step in the reading. A lorry does not raise the room by half a point and stop. It puts water into the air and goes on putting it there while it stands, so a room left alone does not settle — it keeps climbing.

Why this is the vault's real work. Cold and dry both slow the reactions that kill a seed, and they multiply. The freezer is easy: it holds itself. The moisture is the half that has to be held by somebody, every intake week, for the collection's life.

TAKES AS READ

seed is dried to a target moisture before it goes into store

VERDICT 142W

Every arrival here is a step in the rate rather than a step in the reading. A wet lot does not raise the room's moisture by a fixed amount; it adds water to the air and goes on adding it while it stands, so a room left alone climbs and keeps climbing. That is why the dehumidifier has to be moved to a new setting and left there, rather than nudged and released. The second half is the band narrowing. A tenth of a point of overshoot is nothing while the intake is at fourteen per cent, and it is the whole margin once the lots are near six. Both halves are the same lesson the vault is built on: cold and dry multiply, the cold holds itself, and the dry is held by whoever is on the drying floor that week.

TAKEAWAY 13W

A load that keeps arriving is not answered by a correction made once.

3.5 Two parents, and what you are asking of them

HEREDITY & CROSSING · CROSSING HALL · A ROOM · CALLBACK · CHOICE

SCENE 36W

Volpe has two parents pinned above the bench. One is a high-yielding modern variety. The other is a low-yielding landrace that survived the 2018 drought. Otto Bengtsson asks what combination the programme is trying to create.

GUIDE 0W

ASK 11W

What is the programme actually asking for by crossing these two?

TAKES AS READ

offspring get one set of chromosomes from each parent

VERDICT 79W

Meiosis gives each gamete one copy of each chromosome. The F1 therefore receives one set from each parent. The next generation reshuffles those copies into many combinations. The breeder wants offspring carrying useful yield alleles from the modern parent and drought alleles from the landrace. Neither parent has that combination. Many offspring will be worse than either parent because useful and harmful alleles also segregate. Breeding works by making enough recombinant offspring to find the rare combinations worth keeping.

TAKEAWAY 19W

A cross is a request for a combination, and it takes generations to find out whether it was granted.

END OF THE DAY 16W

A population is a set of frequencies, and everything the vault protects is in those numbers.

The season you do not get back

PLAN CARD 104W

Thursday, and the sowing plan is due tomorrow. Volpe's line WM-712 has been through 4 seasons of trials and yields 11% over the standard in dry years. Releasing it needs the whole west ground for seed increase, which is the ground Qureshi's 60 overdue accessions would have used. The forecast for the summer is dry, the merchant wants a decision, and both of them are right about something. Today you work out what a season's delay actually costs, what the trial data does and does not support, and what the crop gets out of a wet spring that a dry one never gives it.

BRIEFING 13W

WM-712 is ready. The ground it needs is the ground the vault wants.

WHAT YOU DECIDE 11W

Weigh a release against a delay, with a dry summer forecast.

4.1 What a season costs

HEREDITY & CROSSING · CROSSING HALL · A ROOM · CHOICE

SCENE 34W

Volpe has the pipeline on the board: cross to release is 7 years, and every stage behind WM-712 moves back with it. Qureshi asks, reasonably, what a year of a wheat variety is worth.

GUIDE 57W

Four accounts of what a season costs. Ask of each whether it is about this line or about the queue behind it. A programme shares seasons, ground and staff. Holding one line delays its seed increase, the trial that would have used the ground, and the crosses waiting. That cost lands on lines nobody was arguing about.

ASK 10W

Why does a one-season deferral cost more than one season?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a breeding programme moves one generation at a time

VERDICT 79W

A breeding programme is a queue that shares seasons, ground and staff. Holding WM-712 for 1 year also delays its seed increase, the trial that would have used that ground, and the crosses waiting behind it. The cost therefore lands on lines that were never part of the argument. That is the strongest version of Volpe's case: delay has opportunity costs throughout the pipeline. It does not prove release is right, but it makes the price of waiting explicit.

TAKEAWAY 15W

The cost of holding something back is rarely paid by the thing being held back.

4.2 What the trial can actually claim

FIELD TRIALS & SELECTION · FIELD LABORATORY · A PERSON · STRESS

SCENE 31W

Four seasons, one site, four replicates each. WM-712 is ahead by 11% in the 2 dry years and by 2% in the wet ones. The merchant's leaflet says "11% higher yielding".

GUIDE 74W

The claim on the leaflet rests on an assumption about how dry next season is, and the slider is that assumption — the soil-moisture deficit, where a low number is a wet year. Three candidate sentences are scored against two criteria. Take the slider down to a wet season and see which sentence is still true, because the one that reads best at the expected deficit is not the one that survives being wrong.

ASK 17W

Which sentence about WM-712 is still true if next season's rainfall lands anywhere in the plausible range?

BEHIND THE BUTTON 167W

What a trial measures. Four replicates at one site over four seasons separate a line from the pieces of ground it was grown on, and they do it under the weather those four seasons happened to bring. The number and its conditions are one object; quoting the number alone makes a different claim.

Why the expected season is the trap. At the four-season average deficit the bare eleven per cent needs the fewest words of qualification, which is exactly what a leaflet optimises for. Take the deficit down to a wet year and the sentence is wrong by a factor of five, and a buyer who paid for eleven and got two has been misled by a figure nobody fabricated.

What survives. A sentence that names the condition it was measured under stays true whatever next season does, and costs the merchant eleven more words on the leaflet. That is the trade being made, and genotype-by-environment interaction is the name for the reason it cannot be avoided.

TAKES AS READ

a trial measures lines under the conditions the trial happened in

VERDICT 142W

The eleven per cent is real and it belongs to two dry seasons. Set next season wet and the same line yields two per cent more than the standard, so a buyer who planned on eleven has been misled by a figure nobody fabricated. The conditional sentence gives up the headline and stays true across the whole rainfall range, which is what robustness means here: not the claim that scores best at the expected value, but the one that is still correct at the edges. The regional claim is worse than both — one site cannot support it at any rainfall, because nothing in the trial varied the region. Genotype-by-environment interaction is exactly this: a line's advantage is a property of the line and the conditions together, and a number quoted without its conditions is a different claim wearing the same digits.

TAKEAWAY 18W

A number from a trial carries the conditions it was measured under, or it is a different claim.

4.3 What the crop cannot do without water

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM · CHAIN

SCENE 33W

Toft has 11 years of yield against summer rainfall and a straight-ish line through it. A trainee asks why a plant that is mostly built from air minds a dry summer at all.

GUIDE 62W

Four steps carry carbon from the air into grain, and each one has a reading beside it saying how much it is passing today. Put them in the order the carbon travels, then name the step that is setting the rate. The largest number on the panel is not automatically the governing one, and neither is the first step in the path.

ASK 21W

Trace the path carbon takes into the grain, and name the step that is setting the rate in a dry summer.

BEHIND THE BUTTON 151W

What a governing link is. Along any path, the step passing the least is the one that fixes the total, and improving any other step changes nothing. That is why naming it matters more than describing the path: it is the only place where an intervention does anything.

Why the stomata are the awkward step. A leaf takes carbon dioxide in through pores and loses water vapour out through the same pores. There is no setting of those pores that admits carbon and holds water, so under drought the plant closes them and accepts the loss of carbon that follows.

What the other steps are doing. Light capture and the enzymes that fix carbon are working at close to their usual rate in a dry summer; the roots are the obvious suspect and are moving water, not carbon. The reading beside each step is what separates a suspect from the answer.

TAKES AS READ

plants take in carbon dioxide through pores in their leaves · photosynthesis as where a crop's dry mass comes from — taken as read

VERDICT 122W

Almost all of a crop's dry mass arrives as carbon dioxide through the stomata, and water vapour leaves through those same pores. Under drought the plant closes them, which is the correct decision for survival and immediately cuts the carbon supply to the sites that fix it. Light interception is unchanged, the fixing enzymes are unimpaired, and the roots are moving what water there is — so improving any of those three changes nothing while the pores are shut. That is what a governing link means: the step passing the least sets the total. The straight-ish line in Toft's eleven years is the same fact plotted at field scale, and the reason breeders talk about water-use efficiency rather than about drought resistance.

TAKEAWAY 15W

The opening a leaf takes carbon in through is the opening it loses water through.

4.4 Why the vault is cold

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · CALLBACK · BALLPARK

SCENE 34W

Mbaye has two lots of the same accession on the bench. 1 spent 4 years at minus 18 degrees and 6% moisture. The other sat in a warm shed and now germinates at 31%.

GUIDE 62W

Deterioration is a chemical rate, so cold and dryness each multiply the time a seed lasts, and the two multiply together. The shed lot is the baseline. Work out how much longer the vault lot should last, using the temperature factor and the moisture factor and nothing else — two of the tiles are quantities from a different part of the problem.

ASK 15W

How many times longer should the vault lot last than the lot in the shed?

BEHIND THE BUTTON 136W

Why a rate rather than a switch. A dry, cold seed is not dormant and not protected; it is still repairing damage and still accumulating it, only slowly. That is why storage life is measured in half-lives of viability and why a seed lot has no shelf date, only a rate.

What the two factors do separately. Around this range each 5 °C drop roughly doubles storage life, and each 2 percentage points of seed moisture removed does much the same. Neither kills anything and neither stops anything — both slow the same reactions.

Why they multiply. The reactions are the same reactions, and each factor divides their rate, so the effects compound rather than add. That is what makes a cold dry vault worth its electricity: sixty-four times is a different institution from eight times.

TAKES AS READ

orthodox seeds are seeds that can survive drying and cold storage · seed viability as a rate: what a stored seed is still doing — taken as read

VERDICT 128W

Cold, dry storage works on a rate. A stored seed is still repairing damage and still taking it on. Both variables slow the same reactions. The shed sat at about 2 °C, with seed at 14 per cent moisture. The vault holds minus eighteen and 6 per cent. That is 20 °C colder. At roughly a doubling per 5 °C, it is 2⁴, or 16 times. It is also 8 points drier. Halve the moisture again and the same thing happens, so that is another few doublings. Both factors divide the rate of one process, so they multiply rather than add. The answer is about 64 times, not 32. That gap is the difference between holding an accession for a working lifetime and holding it for a career.

TAKEAWAY 14W

Cold and low moisture extend orthodox-seed life by slowing the reactions that drive deterioration.

END OF THE DAY 17W

A delay is not neutral, and neither is a release; the question is which cost is recoverable.

What comes with what you wanted

PLAN CARD 94W

Friday, and the plan is signed: WM-712 goes to seed increase, 40 of the 60 accessions get regenerated, and Qureshi has said what she thinks of that in writing. In the glasshouse the backcross lines carrying the resistance gene are all noticeably shorter than their siblings, which nobody selected for and nobody wants. Nakagawa has the marker data and a question about whether the marker still means anything two populations later. Today you work out what is travelling with the gene, whether the marker can be trusted here, and what to do about both.

BRIEFING 11W

The resistant lines are shorter, and nobody bred them that way.

WHAT YOU DECIDE 11W

Find out what else the resistance gene is dragging behind it.

5.1 What came with the gene

HEREDITY & CROSSING · CROSSING HALL · A ROOM · BALLPARK

SCENE 41W

Every backcross line carrying the resistance allele is 12 to 15 centimetres shorter than its susceptible siblings. The donor parent was a short landrace, while the recurrent elite parent is tall. Volpe wants to know what travelled with the target gene.

GUIDE 52W

The scoring sheet counts how often the marker and the height gene come apart. Work out that fraction from the plants that recombined, and read it as a distance along the chromosome. Two tiles belong to the whole population and two to the recombinant class, and only one pairing is the frequency.

ASK 13W

How far apart are the resistance gene and the height locus, in centimorgans?

BEHIND THE BUTTON 156W

Why a distance is the useful form. A recombination frequency of one per cent between two loci is defined as one centimorgan of map distance, so the fraction measured in a population converts straight into how far apart the two genes sit. That is what tells a breeder whether the drag can be broken at all.

Why a chromosome travels in stretches. Crossing over happens at a limited number of points per chromosome per meiosis, so a wanted allele arrives with a block of its neighbours. The closer a neighbour sits, the more generations it takes to separate — and the shortness here is a neighbour, not a cost of resistance.

What the number is for. Six recombinants in two hundred is three centimorgans, which is close enough that finding a line carrying resistance without the height block means screening hundreds of plants rather than dozens. Volpe's answer is therefore a screening budget, not a shrug.

TAKES AS READ

genes sit at fixed places on chromosomes · dominance, and why one allele is enough — taken as read

VERDICT 152W

The short plants are not paying a price for being resistant. They are carrying a block of the donor's chromosome that came across with the resistance allele, and a height gene is inside that block. How often the two come apart is measurable: 6 of the 200 backcross plants carry one without the other, so the recombination frequency is $6 \div 200 = 0.03$, and 1 per cent recombination is defined as 1 centimorgan of map distance. Three centimorgans is close linkage. It means the drag can be broken — the two loci do separate — but only by screening enough plants to catch a 3 per cent event, which is why linkage drag is a budget question rather than a permanent feature of the line. The alternative explanations are all testable and all fail: resistance

costing growth would shorten every resistant plant in every background, and backcrossing does not reduce vigour.

TAKEAWAY 14W

A chromosome is inherited in stretches, so a wanted gene arrives with its neighbours.

5.2 A marker fitted where it worked

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A PERSON · HOLDOUT

SCENE 62W

A marker predicts a trait by sitting close to the gene — in the population it was mapped in. Nakagawa's marker has picked plants correctly for three seasons in that population. She now has a new cross from a different donor, with 100 plants scored both ways. A cut-off's score is the share of plants it calls correctly, so higher is better.

GUIDE 67W

Drag the cut-off and watch how it does on the mapping population. Remember what that number is: the marker was mapped in these plants, so a cut-off tuned on them is tuned to this population's genetics as much as to the trait. When you have chosen, press freeze. Freezing commits you and opens the new cross, which the cut-off has never seen. Report what it scores there.

ASK 36W

Set the marker score above which a seedling is kept, on the plants the marker was mapped in — then freeze it and score it on a population it has never seen. What does it get?

BEHIND THE BUTTON 184W

Why a marker works at all. The marker is not the gene. It is a stretch of sequence near enough to the gene that the two are usually inherited together, so scoring the marker is a cheap way of asking about the gene. How near is near enough depends on the population: parents, generations and any recombination between the two sites.

Why a different donor is a different question. Bring the trait in from another parent and the sequence around the gene comes with it. The distance between marker and gene can be larger, or the marker's variant can travel with the wrong version of the gene entirely. Nothing about the marker has changed; what changed is the population it is being asked about.

What the freeze buys. A cut-off chosen on the mapping plants is flattered by them, and a score measured there is a score of the cut-off plus that population. The held-back cross is the only honest test, and it is only a test while the cut-off was chosen without it — which is what committing before you look actually means.

TAKES AS READ

a marker predicts a trait by being close to the gene, in that population

VERDICT 74W

A cut-off chosen on one population partly fits that population's accidents. Those accidents do not have to repeat. The broad high-scoring shelf is useful because it persists when the rule is frozen and applied to the new donor cross. The sharp optimum at 63 does not persist; it was overfit to the mapping plants. A holdout population separates a marker rule that generalises from one that merely describes the sample used to choose it.

TAKEAWAY 17W

A rule fitted on one population has to be scored on a population it has never seen.

5.3 Short is not simply worse

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM ·
ALLOCATE

SCENE 36W

Toft points out that shorter wheats were the whole point of the 1960s: less stem, more grain, and they stand up in the rain. Volpe still does not want 15 centimetres she did not ask for.

GUIDE 65W

The crop built 19.6 tonnes of dry matter this season and the list wants more than that. Roots and canopy are not optional, so they are already committed. Spend what is left and see which of the questions underneath your plan can still be answered — every one of them is a real thing the station wants, and the plan is what you give up.

ASK 18W

19.6 tonnes of dry matter. Spend it across the plant, and know what the plan stops you claiming.

BEHIND THE BUTTON 160W

What harvest index is. The fraction of everything the crop built that ends up in the grain. A tall wheat puts a quarter of its dry matter into stem; a short one puts less, and the difference is available to the ear. Nothing about the shorter plant photosynthesises faster.

Why the stem is not waste. Straw holds the ear above the ground, carries water and sugars to it, and stops the crop going flat in August rain. A stem thick enough to stand in weather is bought out of the same nineteen tonnes as the grain, which is why the trade is a trade and not an oversight.

Why defence is on the list. Making the compounds that slow a rust infection costs dry matter that would otherwise be grain, and a line that spends nothing on it yields more in a clean year and less in a bad one. One budget, and the season decides which allocation looks clever.

TAKES AS READ

a plant divides what it builds between stem, leaf and grain

VERDICT 145W

A crop cannot build more dry matter than the season allowed it, so every structure is paid for out of one total and the grain is the last claimant. That is harvest index: not how much the plant made, but the share of it that ended in the ear. Roots and canopy are committed — nothing grows without them — which leaves about 9.7 tonnes to divide. Grain fill wants 8.6 of that, a stem strong enough to hold up in August rain wants 5.2, the defence compounds want 0.8, and the extra 15 centimetres of height wants 1.8. They cannot all be bought. Shorter wheats did not photosynthesise faster in the 1960s; they stopped spending as much on stem, and the ear got what the stem gave up. Volpe's 15 centimetres is not a rounding error, it is 1.8 tonnes of somebody else's grain.

TAKEAWAY 16W

What matters is the share of what the plant built that ends up in the grain.

5.4 Why the same line is sown four times

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · CALLBACK · CHOICE

SCENE 37W

Meera Quiroga has the plot map on the wall: 240 plots, with each line appearing 4 times across 4 blocks. The order is randomised, and the north end of the field is visibly wetter than the south.

GUIDE 0W

ASK 10W

What does sowing each line in four randomised plots buy?

TAKES AS READ

the same crop grows differently in different parts of a field

VERDICT 81W

A field varies in drainage, soil depth, compaction and shelter. Sow a line once and its yield contains both genotype and the conditions of that plot. Replicating the line in several blocks spreads it across different ground. Randomising positions prevents the researcher's habits from deciding which line receives favourable places.

Differences among replicate plots estimate environmental variation. That variation becomes part of the uncertainty when lines are compared. Replication therefore improves the genetic comparison rather than the measurement of one plot.

TAKEAWAY 19W

A field is not uniform, so a single plot measures the plot as much as the line in it.

END OF THE DAY 18W

Genes near each other are inherited together, which is what makes markers work and what makes them expensive.

How many is enough

BEFORE THE DAY — HUNT

Six bagged heads, and the sheet lists four

Every bagged head in the crossing block has to be accounted for before the wind gets up, because a bag that has come off is a cross with an unknown father and nothing downstream can recover which one it was. Four are where the sheet says. Find the other two.

PLAN CARD 100W

Monday of the second week, and the regeneration list has to be cut from 60 to 40. Ivanescu has spent the weekend on the arithmetic of how many plants each grow-out needs, and it is not the number in the manual. Mbaye has 11 years of germination tests on 1 accession and a curve that is about to cross his line. Quiroga wants to know why last year's trial found a difference in plant height that this year's cannot see. Today you size the samples, read the curve, and work out which of last year's results were the ground talking.

BRIEFING 11W

40 accessions, 1 field, and a number nobody wants to hear.

WHAT YOU DECIDE 15W

Put a number on what a regeneration needs, and on what the vault is losing.

6.1 Not the number of plants

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A ROOM · BALLPARK

SCENE 35W

Last year's regeneration of a dioecious conservation accession used 120 plants. 18 pollen donors and 102 seed parents contributed to the harvested seed. Ivanescu wants the effective size before the next regeneration plan is approved.

GUIDE 55W

Five numbers, and two of them count plants standing rather than parents contributing: the plot total, and half of it. Ask of each whether this measure uses it. The two parent counts are badly unbalanced, and the answer will come out well below the headcount. Rare alleles had fewer independent routes through the smaller side.

ASK 13W

What effective population size did 18 pollen donors and 102 seed parents produce?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

dioecious means pollen and seed are produced by separate plants · unequal parental contributions reduce effective population size

VERDICT 80W

Effective population size asks how many equally contributing breeders would produce similar genetic drift. For separate pollen and seed parents, $N_e \approx 4N_mN_f/(N_m+N_f)$. Here 18 pollen donors and 102 seed parents give about 61, far below the 120 plants standing in the plot. The imbalance means transmitted alleles had fewer independent routes through the pollen side. Rare variants therefore faced a stronger sampling bottleneck than the headcount suggests. This sex-ratio formula applies to this separate-sex accession, not ordinary selfing wheat.

TAKEAWAY 19W

A regeneration can contain many plants yet behave genetically like a much smaller population when parental contributions are unbalanced.

6.2 Write the viability rule before it falls

CONSERVATION & VIABILITY · SEED VAULT · A PERSON · TRIGGER

SCENE 34W

Mbaye has 11 germination results from the same accession, falling from 98 to 85%. The current regeneration floor is 85. Booking field space takes several planning updates, and starting the grow-out takes one more.

GUIDE 67W

One rule, written before the test results arrive and fixed once released. The grow-out takes one interval from the moment it is called, and below 85 % germination the lot is past the floor the vault regenerates at. So the line sits above 85 by however far viability falls in that interval — and not so far above it that a healthy lot is regenerated for nothing.

ASK 18W

At what germination do you start the regeneration grow-out, given it takes an interval to get under way?

BEHIND THE BUTTON 140W

Why a floor is not a trigger. The regeneration floor is where the accession must already have been grown out. Everything that has to happen first — booking ground, preparing seed, planting — takes time, so the trigger has to sit above the floor by however long that takes.

What the falling series is telling you. Eleven results from 98 down to 85 is a trend, not a scatter, and the rate of fall is what turns a threshold into a date. A rule written against the level alone ignores how fast the level is arriving.

Why the rule is written before the updates. Germination results arrive one at a time, and a threshold adjusted as they come in is a description of what somebody already decided. Committing first is what makes the vault's practice defensible to whoever inherits it.

TAKES AS READ

germination percentage measures the fraction of seeds that can produce seedlings · regeneration requires field space before seed can be multiplied

VERDICT 94W

The viability floor is not the moment to begin acting, because the biological response has lead time of its own. A grow-out starts an interval after it is called, and germination goes on falling in the meantime, so the rule has to sit above the 85 per cent floor by however far viability drops in that interval. Writing it before the record is replayed is what keeps hindsight from becoming policy. In this authored stream, starting at 88 leaves the lot under way before the floor; waiting until 85 acts after it is crossed.

TAKEAWAY 21W

A biological threshold must fire early enough for the action it triggers to arrive before the measured state crosses the limit.

6.3 A difference that was the field

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · CHOICE

SCENE 37W

Last season two pure lines differed by nine centimetres in height, with a statistically clear separation. This season the same lines were randomised onto different plots and no longer differ. The seed stocks themselves have not changed.

GUIDE 53W

Four explanations for a gap that vanished. Ask of each whether the seed or the ground changed between seasons. Lines do not become genetically different in a year. Randomisation prevents systematic favouring and one realised layout can still be lucky. A second season on different plots asks whether the difference follows the seed.

ASK 6W

What is the most likely explanation?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the same line grown in two places can give two different results

VERDICT 80W

A one-season comparison contains genetics plus whatever was unusual about the plots each line received. Randomisation makes systematic favouring less likely, but one realised layout can still be lucky. A second season on different plots asks whether the difference follows the seed. Here the nine-centimetre gap disappears. That makes last year's estimate look partly environmental rather than a stable genetic effect. Replication across plots and seasons is how a breeding programme separates a line from a flattering piece of ground.

TAKEAWAY 16W

A result that does not survive being grown again somewhere else was probably about the somewhere.

6.4 Counting a generation

HEREDITY & CROSSING · CROSSING HALL · A ROOM · CALLBACK · BALLPARK

SCENE 30W

400 F₂ plants, and 306 of them show the resistant reaction while 94 do not. Bengtsson asks what that number is supposed to be, and whether it is close enough.

GUIDE 66W

Work out what a single dominant locus predicts, then compare. Two heterozygous parents give three resistant to one susceptible, so the expected count is the number of plants times three quarters. The two-gene alternative, where either dominant allele is enough, predicts fifteen to one instead. Those two expectations are far apart at four hundred plants, which is what makes the observed count evidence rather than decoration.

ASK 19W

How many of the 400 F₂ plants should show the resistant reaction if a single dominant locus governs it?

BEHIND THE BUTTON 166W

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a quantity belonging to a different part of the problem is caught before it is placed — and two of the tiles here belong to the alternative model, not to this one.

Where the three quarters comes from. Each parent carries one resistant and one susceptible allele, and meiosis puts one of the two into each gamete. Four equally likely combinations follow, and three of them contain at least one resistant allele. Dominance is what makes those three look alike.

Why the comparison needs both models. A count near three hundred is only evidence for one locus because a rival model predicts something else. Fifteen to one puts twenty five susceptible plants in four hundred rather than a hundred, so a single tray of this size separates them. A ratio quoted without the alternative it beats is a number nobody can argue with.

TAKES AS READ

a plant carries two copies of each gene and passes on one

VERDICT 105W

F₁ plants carry one resistance allele and one susceptible allele. Meiosis separates those alleles, so 2 heterozygous parents give a 3 : 1 phenotype expectation when resistance is dominant. In 400 offspring that is $400 \times 3 \div 4 = 300$ resistant and 100 susceptible. The observed 306 and 94 sit close to it. 2 independently assorting genes where either dominant allele gives resistance predict 15 : 1, which is $400 \times 15 \div 16 = 375$ resistant and 25 susceptible. The counts therefore support one dominant locus and rule the two-locus alternative out, which no single plant in the tray could have told you.

TAKEAWAY 15W

The proportions in a whole generation carry information that no single plant in it does.

END OF THE DAY 17W

Every number here is about a population, and populations do not behave like the plants in them.

400 kilometres away

PLAN CARD 96W

Tuesday. The regional bulletin reports a new race of stem rust 400 kilometres east, on a variety that carries the same resistance gene as WM-712 — and it is not stopping it.

Nakagawa can screen for a second gene in the collection, Ivanescu can tell her where to look, and neither of them can do it in one season without giving something up. The line is already in seed increase. Today you decide what is worth screening for, write the rule that says when Wellmere acts, and work out what the crop's own defences cost it.

BRIEFING 14W

A new race of the rust, reported on the other side of the border.

WHAT YOU DECIDE 11W

Decide what to do about a disease that has not arrived.

7.1 One base, one broken lock

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A ROOM · CHOICE

SCENE 37W

WM-712 is failing east of the border. Nakagawa has sequenced a rust gene from older and newer isolates. The newer isolate differs by one nucleotide in a coding region, while the rest of that gene is unchanged.

GUIDE 58W

Four accounts of how one base can matter. Ask of each whether it respects the order of the steps. DNA is transcribed, and the transcript is translated. So a change acts before the protein exists, not during infection. Some substitutions are silent, and not all of them stop translation. What follows is selection on whichever variant escapes recognition.

ASK 11W

How can 1 DNA substitution change whether WM-712 recognizes the pathogen?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

genes contain DNA sequences that can be transcribed into mRNA · ribosomes translate mRNA codons into amino acids · dominance, and why one allele is enough — taken as read

VERDICT 73W

A point mutation replaces one nucleotide in DNA. When that region is transcribed, the change can alter an mRNA codon. During translation, a changed codon may place a different amino acid in the pathogen protein. That can alter the protein's shape or function, although some substitutions are silent. If the altered protein escapes recognition by WM-712, that rust variant can reproduce on resistant plants. Selection then raises the frequency of the resistance-breaking variant.

TAKEAWAY 18W

A mutation matters through what it changes in RNA, protein, and phenotype; selection then acts on that phenotype.

7.2 Where an unused allele lives

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A PERSON · CHOICE

SCENE 32W

Ivanescu has to pick 50 accessions to screen out of 41,000. On the board: modern varieties, the breeding programme's own parents, and the landraces from districts where rust has always been present.

GUIDE 53W

Four kinds of material to search. Ask of each what selection it has actually experienced. A landrace grown where the rust arrives often has met that pathogen many times, so protective alleles could rise in frequency. The programme's own parents may share ancestry, and the same source, with the line that is failing.

ASK 11W

Which material is most likely to carry a different resistance gene?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

populations are shaped by what has been selecting them

VERDICT 85W

Selection leaves a signature in the populations that experienced it. A landrace grown where rust arrives often has repeatedly faced that pathogen. Plants with useful resistance can then contribute more seed, raising the frequencies of protective alleles. Modern breeding parents are more likely to share ancestry with WM-712 and may carry the same resistance source. Landraces from endemic regions therefore offer a better search space for different resistance. They are not guaranteed to contain it, so the next step is still screening rather than assumption.

TAKEAWAY 15W

Where a disease has always been, the plants that stayed are the ones that could.

7.3 Where the missing 2% went

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM · BALANCE

SCENE 32W

Toft has paired no-rust plots of near-identical lines with and without the resistance gene. The resistant line yields about 2% less. Nakagawa also detects the defence transcript in clean leaves before infection.

GUIDE 51W

Read every stream, then count only what belongs in the same resource ledger. The resistant line yields about 2% less in plots with no rust at all, so the resource has gone somewhere other than growth. Nakagawa's transcript measurement is the clue to which stream is not yet on the board.

ASK 14W

Which hidden allocation closes the resistant line's resource budget and matches the yield penalty?

BEHIND THE BUTTON 124W

Why no-rust plots are the whole design. With disease present, a resistant line yields more because it is not being eaten. Removing the disease removes that benefit and leaves only the cost, which is the quantity this stop is trying to measure.

What the transcript is evidence of. The defence pathway is being expressed in clean leaves, before any infection. Making those proteins costs carbon and nitrogen that are then unavailable for grain, and that expenditure happens whether or not the pathogen ever arrives.

Why the number matters commercially. A 2% yield penalty carried permanently is the price of insurance against a disease that may not come every year. Knowing it is what lets a breeding programme decide when the insurance is worth buying.

TAKES AS READ

transcription produces mRNA from DNA · translation uses mRNA to make protein · gene expression can be constitutive or inducible · dominance, and why one allele is enough — taken as read

VERDICT 76W

Regulatory DNA and transcription factors control whether a gene is transcribed. Some genes are expressed continuously, while others are induced by a signal. Here the defence transcript is already present in clean leaves, so the plant is investing in that pathway before rust arrives. Translation turns the mRNA into protein, and maintaining that machinery uses part of the plant's resource budget. The simplified ledger assigns the missing two units to defence, matching the observed yield penalty.

TAKEAWAY 18W

Gene regulation controls when defence proteins are made, and expression before infection can carry a standing resource cost.

7.4 How many plants make a population

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · CALLBACK · CHOICE

SCENE 30W

Mbaye can regenerate the 1994 accession from whatever number of plants the ground allows. Volpe points out that 30 plants would free half a plot. Ivanescu has gone very quiet.

GUIDE 54W

Four things a small grow-out might cost. Ask of each whether careful storage afterwards could repair it. Rare variants are the easiest to miss, because few plants carry them. Once an allele is in none of the contributors, no freezer restores it. And the resulting seed germinates well, which is why nothing routine notices.

ASK 10W

What does regenerating from 30 plants instead of 200 cost?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

each generation's alleles are a sample of the last generation's · genetic drift, and why it is worst in a small sample — taken as read

VERDICT 85W

Every regeneration samples alleles from the population that came before it. Rare variants are easiest to miss because few plants carry them. If an observable type occurs in only a few plants, a small grow-out can contain none of those plants by chance. Once the allele is absent from every contributor, careful storage cannot restore it. The resulting seed may still germinate well and look normal. That is why a genetic bottleneck can damage a collection without leaving an obvious sign in routine viability tests.

TAKEAWAY 18W

A grow-out is a sample, and a small sample loses whatever was rare in what it came from.

END OF THE DAY 17W

A resistance that rests on one gene is a resistance the pathogen only has to solve once.

What you asked for and what you got

PLAN CARD 97W

Wednesday. The trial is scored and Volpe wants to select the best 20 plots out of 240 to carry forward. Quiroga has the numbers for what that will actually produce, which is less than the 20 plots suggest and more than nothing. Bengtsson can give the programme an extra generation in the warm bay if the decision is made this week, and Toft has finally got the canopy data that explains why 2 of those 20 are good. Today you predict the gain from one round of selection, and decide what to do with an extra generation.

BRIEFING 11W

The best 20 plots, and what their offspring will be like.

WHAT YOU DECIDE 9W

Predict what one round of selection will actually deliver.

8.1 What one round returns

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · BALLPARK

SCENE 36W

The trial mean is 7.4 tonnes per hectare. The 20 plots Volpe wants to keep average 9.1. Quiroga estimates yield heritability on this ground at 0.3 and asks what gain should actually reach the next generation.

GUIDE 57W

Five numbers, and two of them belong elsewhere: the number of plots, and a difference already worked out. Ask of each what this prediction needs. The selected plots are above average for two reasons, and only one of them is inherited. So the selected mean is not a forecast; much of its advantage stays with the plots.

ASK 10W

Estimate the yield of the next generation from this selection.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a plant's performance is part inheritance and part where it grew

VERDICT 79W

The selected plots are above average for genetic and environmental reasons. The breeder's equation keeps only the heritable share of that advantage. Here the selection differential is $9.1 - 7.4 = 1.7$ tonnes per hectare. Multiplying by $h^2 = 0.3$ gives a response of 0.51. Adding that response to the original mean predicts about 7.9 tonnes per hectare next generation. The selected-plot mean of 9.1 is therefore not an inherited forecast; much of its advantage stays with the plots.

TAKEAWAY 17W

Only the heritable share of an advantage is passed on, and for yield that share is small.

8.2 An extra generation in the bay

HEREDITY & CROSSING · CROSSING HALL · A PERSON · CHOICE

SCENE 37W

Bengtsson can run one more generation over the winter in the warm bay. It is the only place at Wellmere where a generation costs 3 months instead of 12, and there is bench room for 1 population.

GUIDE 57W

Four uses for one warm bay. Ask of each whether it needs time, field conditions, or space. Yield cannot be judged under glass because the environment is missing. Regenerating a landrace there imposes a selection nobody chose. Seed multiplication needs area. Only one of the four is short of generations rather than of anything the bay lacks.

ASK 10W

What should the warm bay be used for this winter?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

a glasshouse can complete a generation out of season · dominance, and why one allele is enough — taken as read

VERDICT 83W

The warm bay is valuable when calendar time is the main constraint. Breaking linkage drag needs more meioses and more opportunities for a rare crossover, so another generation directly advances that goal. Yield cannot be judged credibly under glass because the field environment is missing. Regenerating a landrace there would impose an unintended glasshouse selection. Seed multiplication mainly needs area, which the bay lacks. The scarce indoor generation therefore belongs to the backcross rather than to work that needs field conditions or space.

TAKEAWAY 11W

A generation indoors is for advancing material, not for judging it.

8.3 Why those two did well

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM · CHOICE

SCENE 34W

2 of the 20 selected plots reached full canopy 11 days before the rest and flowered 4 days earlier. Their yields are the highest in the trial and the season was dry after flowering.

GUIDE 57W

Four explanations for the early lines winning. Ask of each whether it is about efficiency or about timing. Soil water became limiting after flowering, so a canopy that closed earlier caught light while the stomata could still open. Two of these options are about escaping something else entirely. And in a wet year the advantage could vanish.

ASK 10W

Why did the early lines yield best in this season?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

a crop builds most of its grain from light captured after flowering · resource allocation: defence, growth and yield out of one budget — taken as read

VERDICT 79W

Yield depends on how much light the crop intercepts, how efficiently it converts that energy, and how much biomass reaches grain. In this dry season, soil water became limiting after flowering. The lines that closed their canopies earlier captured more light while stomata could still stay open. Earlier flowering also shifted grain filling toward the wetter part of the season. These traits are therefore escape strategies for this environment. In a wet year their advantage could shrink or disappear.

TAKEAWAY 14W

An early canopy captures light while there is still water to use it with.

8.4 One disagreement, two possible causes

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A ROOM · CALLBACK · DEGENERACY

SCENE 40W

Nakagawa has 400 F2 plants scored by a linked PCR marker and by resistance. 11 carry the marker but fail the resistance test. The PCR controls and gel bands look clean; the marker lies about four centimorgans from the gene.

GUIDE 53W

Two controls slide recombination rate and test error along a family of combinations that all produce eleven disagreements in four hundred plants. See how wide that family is before naming a cause. Then bring in the direct-gene test, which fails for one of those explanations and not the other, and watch it collapse.

ASK 16W

Which combinations can explain the first disagreement rate, and which one survives the direct-gene PCR check?

BEHIND THE BUTTON 142W

Why one disagreement rate cannot separate the two. A marker four centimorgans from the gene is expected to be separated from it in about four plants per hundred by recombination. A test that occasionally misreads produces the same disagreements. Eleven in four hundred is consistent with several mixtures of the two.

What the direct-gene test does. It asks about the gene itself rather than about the marker, so recombination between them is irrelevant to it. Plants that carry the gene and failed the resistance test are test error; plants that lack the gene are recombinants.

Why the clean controls are not an answer. Controls show the reaction worked on known material. They cannot detect a plant-specific failure — poor template, an inhibitor, a mislabelled well — which is exactly the kind of error that produces a handful of disagreements in four hundred.

TAKES AS READ

PCR uses primers to amplify a chosen DNA region · gel electrophoresis separates DNA fragments by size · a nearby marker can be separated from a gene by crossing over · independent assortment, and the ratios two genes give instead — taken as read

VERDICT 77W

PCR copies a chosen DNA region through repeated denaturation, primer annealing, and extension. Gel electrophoresis can then separate the resulting DNA fragments by size. Here that tool reads a marker near the resistance gene, not the gene itself. Marker–trait disagreement can therefore come from recombination or assay error. One disagreement rate cannot separate those causes. A second PCR assay aimed

inside the resistance gene adds independent evidence. When it confirms the gene call, linkage and recombination remain.

TAKEAWAY 19W

PCR can identify a DNA sequence, but a linked DNA marker can still separate from the gene it predicts.

END OF THE DAY 18W

What a round of selection returns is the part of the difference that was inherited, and no more.

The collection is not what the records say

BEFORE THE DAY — CANVASS

Which growers actually kept the seed apart

Half the district took seed from Wellmere last year on the understanding it would be grown in isolation, and whether that happened decides what the collection can claim about those lines. Ask round the farms until enough answers have come back to say.

PLAN CARD 91W

Thursday. Nakagawa ran a routine genotyping panel across 90 accessions to build a reference set, and a third of them do not match the passport data — same numbers, same records, different populations. Pirelli has checked the labelling twice and found nothing wrong with it. Qureshi has gone down to the vault and has not come back up. Today you work out what has actually happened to those accessions, whether the vault or the field is where it happened, and what it means for the 50 being screened for rust resistance.

BRIEFING 9W

A genotyping run that was meant to be routine.

WHAT YOU DECIDE 12W

Find out why the regenerated accessions no longer match their own passports.

9.1 Which evidence shares the grow-out

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A ROOM · TRACE

SCENE 34W

30 of 90 accessions no longer match their 1990s allele frequencies. Nakagawa has four evidence channels on the board: genetic comparisons, regeneration histories, germination results and the vault log. Independent shelf checks are clean.

GUIDE 51W

Open each channel to see what it was computed from. Keep the ones that stand independently of the repeated small grow-outs, and untick the ones that inherit that history. Then name the shared source. The shelf checks are clean, which is itself evidence about where the drift did not come from.

ASK 14W

Which results inherit the repeated small-grow-out history, and which evidence remains independent of it?

BEHIND THE BUTTON 126W

What repeated small grow-outs do. Regenerating an accession from a handful of plants each time loses alleles by chance, and doing it repeatedly compounds the loss. The result looks exactly like a collection that has drifted from its 1990s records — because it has, through its own maintenance.

Why the shelf checks being clean matters. If storage conditions or mislabelling were the cause, the shelves would show it. Clean shelves point away from an accident and towards something systematic in how the material has been renewed.

Why some channels survive. Germination results and the vault log are records of handling rather than of allele frequency, so they do not inherit the grow-out history. That is what makes them able to test it rather than repeat it.

TAKES AS READ

a result can depend on earlier handling as well as on the current assay · independent evidence does not share the same critical source · genetic drift, and why it is worst in a small sample — taken as read

VERDICT 80W

Three genetic comparisons depend on the same regeneration history, so their agreement is not three independent causes. The viability panel does not share that path. It rests on germination tests and the vault record, and it says the seed remained alive. Opening the dependencies therefore separates two claims: storage preserved viability, while repeated small grow-outs altered allele frequencies. Discarding the viability evidence would throw away an independent result. Ignoring the shared grow-out path would count one causal exposure several times.

TAKEAWAY 20W

Several results can agree because they share one history; independent evidence should be kept when the shared dependency is corrected.

9.2 What the screen may not find

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A PERSON · CHOICE

SCENE 38W

Nakagawa is screening 50 accessions for a second resistance gene. 22 are on the drifted list, but their marker controls are clean. She wants to know what a negative result from those accessions is still allowed to mean.

GUIDE 60W

Four readings of what drift does to a screen. Ask of each whether it acts on negatives, positives, or both. Drift can remove an uncommon allele, so a clean negative now has two histories and the marker cannot tell them apart. Drift does not invent the target, so positives survive. The remedy is different material rather than a different marker.

ASK 8W

What does the drift mean for the screen?

BEHIND THE BUTTON 195W

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TAKES AS READ

an allele that is uncommon in a population may be missing from a small sample of it

VERDICT 81W

Drift can remove an allele that was uncommon in the original population. A clean negative from a drifted accession therefore has two explanations: the allele was never present, or it was present and disappeared during regeneration. The marker cannot distinguish those histories. A positive remains useful because drift does not invent the target allele. The remedy is not a different marker. It is to test material that has avoided the bottleneck, or to compare with older archived material when that exists.

TAKEAWAY 22W

A screen finds what is in front of it, and what is in front of it is what survived the last grow-out.

9.3 What the store did right

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · ATTEST

SCENE 38W

Qureshi wants one sentence for the board that names the failed process rather than the room it happened near. The vault temperature log covers 28 years without an excursion, and current germination remains strong in the affected packets.

GUIDE 65W

Five signed statements, and three verifications. Every one of them is somebody's record rather than a condition you have observed, so decide which two would hurt most if the paper were wrong, and check those. A claim can only be held once it has been verified. Holding the whole critical list is not caution, because two of these are backed by evidence taken this week.

ASK 14W

Which of these signed claims cannot be held, given what the evidence actually supports?

BEHIND THE BUTTON 155W

What a signature records. It records who is answerable for a statement, and nothing about whether the statement is true. The two come apart exactly where nobody looked, which is why the verifications are fewer than the claims — an audit with time for everything measures nothing about where you chose to look.

Why the freezer is the wrong suspect. Cold storage slows deterioration, and a continuous logger trace with calibration certificates behind it is about as good as institutional evidence gets. Germination in the affected packets is strong, which is what a working freezer looks like from the outside.

What a grow-out record has to say. A regeneration is a sampling step, so the two things that decide whether it preserved the accession are how many plants contributed and whether they were isolated from foreign pollen. Both are written down by the person who did the work, on the day, or not at all.

TAKES AS READ

a store preserves what it is given rather than improving it

VERDICT 120W

The freezer claim is critical and backed — 28 years of continuous logger trace, calibrated, with strong germination in the affected packets to match. The two grow-out claims are critical and unbacked, and they are where a collection actually loses alleles: a regeneration is a sampling step, so a missing plant count means nobody can say how large the sample was, and a missing isolation record means nobody can say whose pollen it received. Both were signed by people who were doing the work correctly as far as anyone knows, which is the point. A signature records who is answerable, not what happened. Naming this a storage failure sends money to the one piece of equipment that did its job.

TAKEAWAY 17W

Storage held what it was handed; the loss happened in the field, on the way back in.

9.4 Where a tonne of grain comes from

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM ·
CALLBACK · BALANCE

SCENE 36W

Toft has 11 tonnes from 1 field on the weighbridge slip and a trainee who has just said the crop got it from the soil. He would like that answered properly before it goes any further.

GUIDE 62W

Close the ledger. The rows are what the field took up or gave off over the season, and the total is the eleven tonnes of dry matter on the slip. Count the ones that are actually part of that dry weight and leave out the one that is a different quantity altogether. Reading a row costs nothing; counting it is the claim.

ASK 23W

Close the season's dry-matter ledger for the field. Which flows make up the 11 tonnes, and what is the term nobody wrote down?

BEHIND THE BUTTON 151W

What dry matter is. Grain is weighed at fifteen per cent moisture and the ledger is kept dry, so the water in the standing crop is not in the total at all. About nine tenths of what remains is carbohydrate built from carbon dioxide.

Why one row cannot be counted. The season's transpiration is four thousand tonnes of water, and it is not a component of dry mass — it is the flow that carried the minerals up and kept the leaves cool. Adding it to a mass ledger gives a number two orders of magnitude wrong and no warning.

What the minerals do. Nitrogen, phosphorus and potassium decide how much tissue the crop can build and contribute a few per cent of its weight. That is the distinction the trainee is missing: what controls the size of the harvest is not the same as what the harvest is made of.

TAKES AS READ

plants build sugars from carbon dioxide and water using light

VERDICT 143W

The eleven tonnes on the slip came out of the air. Photosynthesis fixed about 19.6 tonnes of dry matter over the season, respiration burned 6.1 of it back to carbon dioxide, and minerals from the soil account for 0.3 — a few per cent, which is why the trainee's answer sounds plausible and is wrong by an order of magnitude. The ledger still does not close, because 2.4 tonnes of it is root and stubble that stayed in the field: harvest index is the fraction of what the crop built that leaves in the trailer, and it is never one. The transpired water is the row that cannot be counted at all. Four thousand tonnes of it moved through the crop, and none of it is dry mass — it is the flow that carried the minerals and kept the leaves from cooking.

TAKEAWAY 16W

A harvest weighs what it weighs because of something that was not there when it started.

END OF THE DAY 14W

A collection can be kept perfectly and still stop being the thing it was.

What is left to work with

PLAN CARD 97W

Friday. The finding is written up and the season has to be re-planned around it. 22 of the 50 screening candidates are unreliable, the ground is already committed, and the 1 source of unbottlenecked material is Idris Fenn's 4 landraces, which have been grown every year on his own farm and selected by his own weather. Fenn is willing. Volpe points out, not unfairly, that none of this changes what is already in seed increase. Today you decide what the screen is run on, what regeneration looks like from now on, and what the field can absorb.

BRIEFING 15W

The vault is one copy. The other 1 is in a field 40 miles away.

WHAT YOU DECIDE 13W

Rebuild the season's plan around material that has not been through a bottleneck.

10.1 The other copy

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A ROOM · CHOICE

SCENE 38W

Fenn's 4 landraces descend from populations sampled for Wellmere in 1994. His family has grown them every year since then, usually as large fields. The vault copies have been regenerated only twice, each time through much smaller grow-outs.

GUIDE 50W

Four things the farm copy might have. Ask of each whether it is a claim about age or about population size. Fenn's seed is not unchanged since 1994: his soil, weather and disease have selected it every year. What it avoided is a grow-out of dozens standing in for thousands.

ASK 10W

What does Fenn's seed have that the vault's does not?

BEHIND THE BUTTON 195W

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TAKES AS READ

a population in cultivation continues to be selected by where it is grown

VERDICT 78W

Fenn's seed is not unchanged since 1994. His soil, weather, disease and seed saving have selected it every year. What it has avoided is the vault's small-sample bottleneck: a field contains thousands of contributing plants instead of dozens. Rare alleles lost by drift in the stored copy may therefore persist on the farm. The two conservation strategies preserve different things. Ex situ storage slows evolutionary change; in situ cultivation keeps a large population evolving under its local environment.

TAKEAWAY 12W

Material that never stopped growing has never been through a sampling step.

10.2 What a grow-out has to be from now on

CONSERVATION & VIABILITY · SEED VAULT · A PERSON · SEQUENCE

SCENE 30W

Qureshi has to write a standard that survives the next 20 years and the next 4 season leads. The ground available does not change and the collection does not shrink.

GUIDE 62W

All four clauses will be in the standard, so the order is about what cannot be recovered. Ask of each whether later care could put it back. An allele never sampled into the new seed is gone whatever else is done well. Pollen from next door can be tested for afterwards. And a record protects the next curator rather than this seed.

ASK 11W

Order what a regeneration standard has to fix, most binding first.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

the effective size of a grow-out depends on how many plants contribute

VERDICT 86W

The minimum contributing population comes first because no later care can restore an allele that was never sampled into the new seed. Isolation then prevents pollen from another accession entering the grow-out. Even harvesting protects the effective size promised by the plant count; a few prolific plants should not supply most of the seed. Finally, the record preserves what actually happened for the next curator. A durable regeneration standard therefore controls population size, gene flow, contribution balance and provenance rather than only producing seed that germinates.

TAKEAWAY 17W

A standard is what survives the person who wrote it, so it has to name a number.

10.3 Ground that is already committed

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · SCIENCETANK

SCENE 30W

Every plot is allocated. Quiroga can take the guard rows out of the yield trial, shorten the plots, or drop the fourth replicate. Each buys ground and costs something different.

GUIDE 98W

The yield trial is 240 plots: 60 lines in four replicates. Guard rows take about 8 per cent of the area and stop plots borrowing light and water from their neighbours. Shortening every plot by a fifth frees 20 per cent and raises the error variance, because a short plot samples less ground. Dropping the fourth replicate frees a quarter and removes most of the trial's power to separate lines that differ by under half a tonne. The screening material needs the equivalent of 30 plots, and the headlands and the spare block are outside the measured trial.

ASK 13W

Spend the ground you can free without spending the trial's ability to answer.

BEHIND THE BUTTON 193W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says what you think is likely, what is worth hedging against, and what is not worth pursuing at all — and the last two are where most of the information is. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. That is why the small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead: below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under a line of work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution; they are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

a trial's precision depends on its replication and its plot size

VERDICT 86W

The first place to find ground is outside the measured yield trial. Headlands and spare block can hold a screening nursery that mainly needs many plants and reliable inoculation. If more ground is still needed, shortening every plot spreads a modest precision cost across the trial. Dropping a replicate removes an entire independent comparison. That weakens small yield differences. Removing guard rows is worse because vigorous plots can borrow light and water from neighbours, quietly biasing every comparison. Different cuts therefore sacrifice different kinds of evidence.

TAKEAWAY 19W

Ground taken from a trial is taken from what the trial can say, and which part is a choice.

10.4 What a season costs

HEREDITY & CROSSING · CROSSING HALL · A ROOM · CALLBACK · CHOICE

SCENE 34W

Volpe has the pipeline on the board: cross to release is 7 years, and every stage behind WM-712 moves back with it. Qureshi asks, reasonably, what a year of a wheat variety is worth.

GUIDE 57W

Four accounts of what a season costs. Ask of each whether it is about this line or about the queue behind it. A programme shares seasons, ground and staff. Holding one line delays its seed increase, the trial that would have used the ground, and the crosses waiting. That cost lands on lines nobody was arguing about.

ASK 10W

Why does a one-season deferral cost more than one season?

BEHIND THE BUTTON 195W

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TAKES AS READ

a breeding programme moves one generation at a time

VERDICT 79W

A breeding programme is a queue that shares seasons, ground and staff. Holding WM-712 for 1 year also delays its seed increase, the trial that would have used that ground, and the crosses waiting behind it. The cost therefore lands on lines that were never part of the argument. That is the strongest version of Volpe's case: delay has opportunity costs throughout the pipeline. It does not prove release is right, but it makes the price of waiting explicit.

TAKEAWAY 15W

The cost of holding something back is rarely paid by the thing being held back.

END OF THE DAY 14W

A collection kept in one place is one copy, however carefully it is kept.

A week when the work is just the work

PLAN CARD 119W

Monday of the third week, and for once there is nothing to argue about. The trial is drilled, the screening nursery is on the headland, Fenn's four landraces went in on Saturday and the forecast is a fortnight of ordinary weather, with cloud sitting on the ridge north of the shelterbelt and nothing else in the sky. Wren Ashworth has the vault door open and a backlog of packets to re-bag. Toft is out with the canopy meter because this is the week the readings are worth taking. Today you spend a quiet week on the things that are impossible in a busy one — and on the one measurement that only works while a crop is still growing.

BRIEFING 11W

The drill is in the field and the forecast is kind.

WHAT YOU DECIDE 15W

Sow the ground, and do the things that only get done when nothing is wrong.

11.1 Measuring while it is still happening

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM ·
VALUE

SCENE 39W

Toft can measure canopy cover this week, while every plot is still green, or wait for yield in August. The board already records flowering date and soil-water status, but several canopy measurements are about to disappear with leaf senescence.

GUIDE 60W

Two days of somebody's time, and a list of measurements that cost more than that between them. Buy the ones that would change what the station does, not the ones that would be interesting to have. Two of these can only be taken this week; the others will still be available in August, which is a different kind of cost.

ASK 18W

Two days of field time this week. Which measurements are worth taking, and which would only be interesting?

BEHIND THE BUTTON 143W

What makes a measurement decisive. Not how precise it is, and not how much it tells you — whether the answer changes a decision. A reading that comes out either way and leaves the same next step is worth buying last, however good the instrument.

Why the same axis twice is the mistake. Two measurements of the growing canopy tell you nearly the same thing, so the second one buys precision rather than information. Spending the whole two days on one axis is how a season ends with a beautifully characterised canopy and no idea what the roots did.

Why some of it cannot wait. Canopy cover and green leaf area stop existing at senescence, and no August measurement reconstructs them. Yield can be measured later by definition. That asymmetry, not the relative interest of the numbers, is what should decide this week.

TAKES AS READ

a crop's final yield is built up over the whole season · resource allocation: defence, growth and yield out of one budget — taken as read

VERDICT 122W

A measurement earns its cost by changing a decision. Canopy cover during grain fill and a root core at depth answer different halves of one question — whether WM-712's dry-season advantage comes from intercepting more light or from reaching more water — and the answer decides which trait the programme selects for next season. Both disappear with senescence. Grain moisture and thousand-grain weight are real numbers that will still be available in August, so buying them this week spends a scarce day on an abundant measurement. And a second canopy index buys precision on an axis already covered, which is the commonest way a measurement

budget is wasted: two readings of the same thing feel like twice the evidence and are not.

TAKEAWAY 17W

A measurement taken after the fact records an outcome; one taken during it can explain the outcome.

11.2 Three jobs, one still damaging seed

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · DELEGATE

SCENE 37W

Ashworth has 400 packets with failed foil seals, a viability-testing schedule 3 months behind, and a stack of faded labels still recoverable from shelf positions. The vault is quiet this week, and three staff roles are available.

GUIDE 53W

Three jobs, three staff roles, and one you keep yourself. For each handoff, name the owner, the first real action, and what brings the problem back to you. One of these three is still damaging seed while you decide, which is what should decide the order rather than how large each backlog is.

ASK 14W

Which problem stays with you, and what complete handoffs do the other two receive?

BEHIND THE BUTTON 125W

Which one is getting worse. Four hundred failed foil seals are letting moisture into seed right now, and every day of that is viability that cannot be recovered. The testing backlog is three months old and stable. Faded labels are recoverable from shelf positions and will still be recoverable next week.

Why a first action is required. 'Sort out the seals' is a wish. 'Reseal in batches of fifty, starting with the accessions with no duplicate elsewhere' is a task somebody can begin without asking you anything.

What a return condition looks like here. Not a status meeting. A rule: any packet whose seed shows visible moisture damage comes to you rather than being resealed, because that is a viability decision rather than a repair.

TAKES AS READ

moisture exposure accelerates deterioration of stored orthodox seed · a viability test measures condition but does not repair it

VERDICT 81W

A failed seal is the only problem that is actively changing the seed. Moisture can enter now, and ageing caused by that exposure cannot be undone later. The viability backlog is loud but stable; it delays knowledge rather than causing the decline it measures. Faded labels are also stable while shelf positions still recover identity. The season lead should therefore keep the failed-seal response and delegate the other work completely. Delegation protects attention without turning the quieter jobs into forgotten work.

TAKEAWAY 23W

Urgency depends on whether damage is still accumulating and reversible; a handoff is complete only with an owner, first action and return condition.

11.3 The fortnight when it is possible

HEREDITY & CROSSING · CROSSING HALL · A ROOM · SEQUENCE

SCENE 31W

Crossing needs the female parent before the anthers shed and the male in flower on the same morning. Bengtsson has the 2 parents at the right stage for about 11 days.

GUIDE 56W

All four steps will happen inside one fortnight, and flower biology forces the order. Ask of each what would already be too late. Anthers shed pollen, so a head left open can pollinate itself before the intended cross begins. And the label is not clerical: without it the seed grows perfectly and nobody knows the father.

ASK 10W

Order what has to happen to make a controlled cross.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a plant can only be crossed while its flowers are at the right stage

VERDICT 82W

The order is forced by flower biology. Anthers must be removed before they shed pollen, or the female can self-pollinate before the intended cross begins. The head is bagged immediately so stray pollen cannot enter. Pollen from the chosen male is collected when it is ready, transferred to the receptive female, and the head is re-bagged. The label then records both parents and the date. Without that provenance, seed may grow perfectly while the breeder no longer knows who its father was.

TAKEAWAY 17W

Some work has a window rather than a deadline, and the window is set by the plant.

11.4 Bagged, or open to the wind

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A PERSON · BELT

SCENE 33W

The threshing floor has two hours of bags to clear and one store manager. A bagged head was pollinated by itself under a paper cover. An open head took whatever the wind brought.

GUIDE 45W

Two bins, and the question is what fathered the seed in the bag. A head that was bagged before it flowered can only have pollinated itself. A head left open took pollen from whatever was flowering upwind. Sort on that, not on the label wording.

ASK 12W

Send each bag to the bin that says what fathered its seed.

BEHIND THE BUTTON 89W

Why it matters at the store door. Selfed seed is a known quantity: its two copies came from one plant, so a line stays a line. Open-pollinated seed carries an unknown father and cannot be treated as the same accession, however good it looks in the bag.

Why the wording will not do it for you. Both bins contain bags marked with the same words — covers, sleeves, sheets and skirts appear on both sides — because a cover put on after flowering is not a cover at all.

TAKES AS READ

a flower can be pollinated by its own plant or by another one

VERDICT 123W

The bin a bag belongs in is decided by one moment: whether the head was isolated before its flowers opened, not whether anything was over it at harvest. A bag put on the day before flowering keeps the line — the two copies in every seed came from one plant, and next season's row will be what this row was. A head that was open at flowering took pollen from whatever was upwind, which at Wellmere means the trial plots eighty metres away, and no cover applied afterwards changes it. That is why the words on the labels are no help and are not meant to be: covers, sleeves and sheets appear on both sides of the floor. The date is the category.

TAKEAWAY 16W

What fathered a seed is decided before the flower opens, not by what the bag says.

11.5 Frequencies, not plants

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A ROOM · CALLBACK · BALLPARK

SCENE 36W

Ivanescu has 100 plants from the 1994 accession scored at 1 locus. 36 are AA, 48 are Aa and 16 are aa. He asks for the allele frequency, then points to Hardy–Weinberg as the comparison model.

GUIDE 54W

Five numbers, and two of them count plants rather than allele copies: the sample size, and the recessive homozygotes. Ask of each whether this frequency needs it. A hundred diploid plants carry two hundred copies, and a heterozygote contributes one of them. Genotype counts and allele counts are different arithmetic on the same plants.

ASK 10W

Estimate the frequency of the tall allele in this accession.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a diploid plant carries two copies of the locus · Hardy–Weinberg is a null model that predicts genotype frequencies from allele frequencies

VERDICT 72W

100 diploid plants carry 200 allele copies. The 36 AA plants contribute 72 A copies, and the 48 heterozygotes contribute 48 more. That gives $p = 120/200 = 0.60$ and $q = 0.40$. Hardy–Weinberg predicts $p^2 = 0.36$, $2pq = 0.48$ and $q^2 = 0.16$. Those values exactly match the observed genotype fractions here. This gives a clean null model before drift, selection, migration, non-random mating or a bottleneck changes the population.

TAKEAWAY 15W

A frequency describes the population; a genotype describes one plant, and they answer different questions.

END OF THE DAY 14W

A quiet week is when the work that has no deadline finally gets done.

The rust arrives

BEFORE THE DAY — EVADE

The merchant who wants a release agreed at the gate

The merchant has been on the causeway since eight wanting a yes on WM-712 before the rust question is settled, and Qureshi, the collection curator, has said plainly that nothing is agreed outside the office. Keep clear of him and finish your round.

PLAN CARD 102W

Tuesday. The wind has come round off the foothills and the new race is 200 kilometres closer than it was on Friday and there are pustules on the screening nursery this morning — the same nursery WM-712's sister lines are in. The single resistance gene is not holding. Volpe, who was right in March about the season, has stopped arguing. Nakagawa's screen of the 50 candidate accessions has come back with 2 hits, and Ivanescu can tell you what is wrong with both of them. Today you find out what the collection can still offer, and where the rest of it is.

BRIEFING 11W

200 kilometres in 4 days, and WM-712 is in the ground.

WHAT YOU DECIDE 14W

Find a second resistance gene, in a collection that may no longer have one.

12.1 Which positive is worth the screenhouse slot

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A ROOM · VALUE

SCENE 40W

2 of 50 accessions carry a different resistance marker, and the screenhouse can increase only 1 this season. One seed vial is visibly thin and the other is full. Nakagawa has records and tests available before the slot is committed.

GUIDE 55W

Two positives, one screenhouse slot, and records and tests available before you commit. Open each piece of evidence and ask what it would change about which accession to increase. One vial is visibly thin and the other is full, which is a fact about how much seed there is rather than about which marker matters.

ASK 11W

Which evidence is worth buying before choosing the accession to increase?

BEHIND THE BUTTON 128W

Why the thin vial is the awkward one. Increasing an accession that has little seed left is the most valuable thing the screenhouse can do, because that material is closest to being lost. It is also the riskiest, since a failed increase spends the last of it.

What evidence would actually change the choice. Whether the marker in each accession is the one that confers resistance, whether the thin vial's seed is still viable, whether either accession is duplicated elsewhere. Any of those can flip the decision; a repeat of the marker screen cannot.

Why buying evidence beats choosing carefully. The slot is committed for the season, and a wrong choice cannot be corrected until next year. That asymmetry is what makes a test worth its cost here.

TAKES AS READ

a regeneration can change allele frequencies by sampling · a small viable seed lot can sometimes be increased without first using it in a field trial

VERDICT 85W

The decision is not which accession has more seed today. Seed quantity can be repaired by one careful increase if enough viable seed remains. The harder issue is genetic integrity. The regeneration-history audit shows that one positive has never passed through a grow-out. The abundant one was regenerated three times and belongs to the drifted group. A repeat marker or small yield test may add detail, but neither restores the original population. Evidence is valuable when it changes which source deserves the scarce increase slot.

TAKEAWAY 20W

The best measurement is the one that can change the decision, not the one that produces the most detailed description.

12.2 What was lost, and when

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A PERSON · DIAGNOSIS

SCENE 31W

Ivanescu has the 1994 field notes for the drifted accession. The collector recorded rust on the crop and noted that about 1 plant in 20 stayed clean. Today's sample has none.

GUIDE 53W

Five readings, and two of them are absences of selection: clean nurseries and small contributor counts. Ask of each candidate how many of the five it fits. Five per cent of plants is uncommon, and a sample of thirty misses an uncommon type surprisingly often. Three grow-outs are three chances to lose it.

ASK 8W

What happened to the resistance in this accession?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

an allele present at low frequency can be lost when a population is sampled

VERDICT 76W

The 1994 field note says about 5% of plants showed the resistant phenotype. That is not automatically a five-per-cent allele frequency. Dominance and genotype frequencies matter. It still tells us the resistant type was uncommon. If 5% of plants show it, a random sample of 30 has about a 21% chance of missing it entirely. Repeated small grow-outs create repeated opportunities for that loss. The clean nurseries provide no selection that would reliably preserve the resistance.

TAKEAWAY 17W

An allele that has vanished was not necessarily destroyed; it may only have failed to be picked.

12.3 The copy that was never in the vault

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · CHOICE

SCENE 43W

Fenn's landraces are established in the station plots. His family has grown one of them continuously since before Wellmere existed, in a district where rust arrives most years. The vault copy of the same old population has been regenerated only in clean nurseries.

GUIDE 53W

Four reasons the field copy might hold what the vault lost. Ask of each whether it names a mechanism. Freshness and quantity are not mechanisms. Rust arriving every year means plants with useful resistance leave more seed, which is selection. The clean nursery gave resistance no advantage at all, and rare variants drifted.

ASK 11W

Why is Fenn's field likely to hold what the vault lost?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a population growing where a disease occurs is being selected by it every season

VERDICT 81W

Fenn's population has faced rust repeatedly, so plants carrying useful resistance can leave more seed in infected years. That changes allele frequencies through natural and farmer-mediated selection. The clean vault grow-outs did something different: without rust, resistance had no selective advantage and rare variants could disappear by drift. The farm population is not a frozen copy of 1994, but it may retain or enrich alleles the ex situ copy lost. In situ and ex situ conservation therefore preserve different evolutionary histories.

TAKEAWAY 18W

A population kept in cultivation keeps being tested by the thing you might one day need it for.

12.4 Move the line, not the story

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · CALLBACK · CONTROL

SCENE 38W

Quiroga has three microplot positions across the slope and enough seed from two numbered lines for a quick repeat. The trainee's strip layout put each line in only one place. Yield scatter from repeated microplots is already known.

GUIDE 68W

The number you are watching is plot yield, and repeated microplots already tell you how much it scatters on its own. Three things about the setup can be changed: which line is sown, where on the slope it sits, and how thickly. A change counts only if it moves yield by more than that scatter. Change one, run the microplot, restore it, and name the variable yield follows.

ASK 17W

Which variable moves yield by more than the repeat scatter and then returns when you restore it?

BEHIND THE BUTTON 166W

Why the trainee's layout cannot answer the question. Each line was sown in one place, so line and position changed together. If the numbers differ, nothing in the data can say whether the better line or the better ground did it. That is what a confound is, and it is not fixed by more careful measurement of the same plots.

What the repeat scatter is for. Sow the same line twice in the same conditions and the two plots do not match: soil varies, weather varies, a hare gets in. That spread is the yardstick. A difference smaller than it is a measurement of the field rather than of the treatment.

Why position is the one to suspect on a slope. Water, depth and warmth all change down a slope, and all three move yield. A field trial is laid out to break the link between treatment and position on purpose — blocks, randomisation, replication — and this stop is what happens when that was skipped.

TAKES AS READ

field position can change crop yield · a causal comparison changes one candidate cause while holding the others fixed · dominance, and why one allele is enough — taken as read

VERDICT 81W

A strip design changes genotype and ground together, so either can explain the yield difference. The control run holds the line and sowing density fixed while moving only the plot position. That change shifts yield far beyond the repeat scatter. Restoring the original position restores the reading. Changing the line on the same ground produces only a small shift in this authored test. The result shows why blocks and randomisation are needed before genotype can be credited for a field difference.

TAKEAWAY 23W

When genotype and field position always change together, their effects are confounded; a control test separates them by changing one at a time.

END OF THE DAY 20W

The insurance was bought decades ago by people who did not know what for, and only some of it survived.

Three things before the drill

PLAN CARD 92W

Wednesday, and three things need deciding before the drill goes out at two. The screening nursery is infected and Nakagawa can score it today or lose the reading to fungicide drift from the neighbouring farm. The merchant wants WM-712's release confirmed or cancelled by 5, with 40 tonnes of seed already increased. And Fenn has rung to say he has rust in the field the landraces are in, which is either a disaster or the most useful thing that has happened all season. Today you decide the order and live with it.

BRIEFING 15W

The rust is here, the merchant wants an answer, and the seed increase is halfway.

WHAT YOU DECIDE 9W

Rank a nursery, a merchant and a phone call.

13.1 What cannot be done later

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · SEQUENCE

SCENE 41W

The nursery can be scored this afternoon or not at all because neighbouring fungicide drift reaches it tomorrow. The merchant wants an answer by five. Fenn's infected field will still be there tomorrow, and the seed-increase arithmetic can be done later.

GUIDE 62W

Four jobs, and the rail is not a clock. Order them by how soon each one stops being possible at all, soonest first. Every one of them is available right now; what separates them is which ones will still be available tomorrow, and a job that can be done any time belongs at the end however loudly it is being asked for.

ASK 11W

Order the four jobs by how soon each stops being possible.

BEHIND THE BUTTON 138W

Why urgency and irreversibility are different. The merchant's five o'clock is a deadline somebody set and can move. Fungicide drift arriving on the nursery is a deadline the weather set, and after it the scores that would have distinguished the resistant seedlings do not exist to be taken.

What a lost measurement costs. A drifted nursery does not give a wrong score, it gives a clean-looking one — every seedling protected, no discrimination between them, and a season's screening spent. Nothing later recovers it, because the plants were the instrument.

Why the arithmetic goes last. A seed-increase calculation uses numbers that are already recorded and will still be recorded tomorrow. It feels productive and it is the only item on this list with no expiry, which is exactly why it is the one to do in the evening.

TAKES AS READ

some measurements are only available while conditions last

VERDICT 141W

The nursery stops being an instrument the moment the drift arrives, and it stops silently — every seedling looks protected, so the screen returns a clean sheet rather than an error. That makes it first. Fenn's block is the same kind of asset with a longer fuse: the rust is active for days rather than hours, but once the infection has run its course the plants that stayed clean cannot be told from the plants that were never tested. The merchant's five o'clock is a deadline a person set, and a person can move it; the release decision loses nothing by being made at six. The seed-increase arithmetic uses numbers already in the book and will be exactly as available next week. The loudest item on the list is third, and the two that would have been quietly lost are first.

TAKEAWAY 12W

What goes first is what stops being possible, not what is loudest.

13.2 40 tonnes and a failing gene

HEREDITY & CROSSING · CROSSING HALL · A ROOM · CHOICE

SCENE 32W

40 tonnes of WM-712 seed exists. Its single resistance gene is being beaten 200 kilometres away and in Wellmere's own nursery this morning. It still yields 11% more in a dry year.

GUIDE 56W

Four things to say to the merchant. Ask of each what it does to the forty tonnes that already exist. Destroying seed does not repair a gene, and no statement changes the genetics of what is in the shed. The risk is a variety sold as resistant being planted as though one failing gene were enough.

ASK 6W

What should Wellmere tell the merchant?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a released variety is planted over a large area at once

VERDICT 82W

The dry-year yield advantage is real, and 40 tonnes of seed already exists. Destroying it does not repair the resistance gene. The risk is that a variety sold as resistant may be planted widely as though one failing gene were enough. Naming the weakness changes management and surveillance. Adding a second gene belongs in the next breeding cycle because existing seed cannot be genetically changed today. The biology defines the trade-off, while release policy still sets what level of risk is acceptable.

TAKEAWAY 14W

What makes a variety valuable is also what makes a single failing gene expensive.

13.3 An outbreak in the right place

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A ROOM · CONTROL

SCENE 40W

Fenn's landraces are carrying rust in the station plots, and he asks whether the whole block should be sprayed. Ivanescu has flagging tape, paper bags and plot tags ready, because some individual plants still look clean among heavily infected neighbours.

GUIDE 71W

Four things differ between the clean plants and the infected ones, and the scoring bench will read a rust severity for whatever you set up. Change one of them and run it; changing several at once tells you nothing about any of them. Then put your suspect back the way it was and run it again, because a difference that does not follow the change both ways was never the cause.

ASK 15W

Which difference between the clean plants and their infected neighbours is actually keeping them clean?

BEHIND THE BUTTON 153W

Why the field and not the glasshouse. Resistance that matters is resistance to the pathogen races actually present, at the density they actually reach. A field with the disease in it applies that test to every plant at once and for nothing, which is why the clean plants among infected neighbours are the interesting ones.

Why one thing at a time. Any two plots differ in dozens of ways. A trial that moves several of them cannot say which one the reading followed, and reports itself as ambiguous rather than as a finding — the honest outcome, and the one that costs a season.

Why reversing it matters. Restoring the suspect and watching the reading come back is what separates a cause from a coincidence that happened during the same fortnight. Spraying the whole block removes the test before it has been run, and the plants that would have identified themselves stay anonymous.

TAKES AS READ

a population under a disease is being selected by it

VERDICT 121W

Four things could explain a clean plant sitting among infected ones: its own genotype, the row's exposure to wind-blown spores, the fungicide overspray drifting from the neighbouring trial, and how much nitrogen that strip received. Only one of them survives being isolated. Matching position, drift and nitrogen and changing the seed alone moves the severity score from 48 to 6, and putting the susceptible seed back brings it to 44 — a response that follows the change in both directions, and eight times the trial-to-trial noise. Spraying the whole block would have removed the difference the station most needed to see. This is what standing variation means: the useful alleles were already there, and the outbreak is what made them visible.

TAKEAWAY 16W

A field with the disease in it is the only place the useful plants identify themselves.

13.4 What the rogueing crew is pulling today

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A PERSON · SPOT

SCENE 38W

The rogueing crew works to one standing instruction, chalked at the end of the row. It is rewritten when the season turns, by whoever is running the nursery that morning, and nobody walks the rows to announce it.

GUIDE 53W

Pull the plants the instruction currently wants and leave the rest. The chalked rule changes during the morning and nothing marks the moment. What is scored is the plants either side of a change, because they are the only ones that show whether the crew is reading the board or working from memory.

ASK 12W

Pull to the instruction on the board, and keep watching the board.

BEHIND THE BUTTON 102W

Why the instruction changes at all. Early in the season the nursery is being cleaned of anything that is not the line — off-types, volunteers, the wrong height. Once rust arrives the same crew is looking for something else entirely, and the plants that were being pulled an hour ago are now the ones worth keeping.

Why the changeover is where the damage happens. A crew that keeps pulling to the old rule for ten more minutes removes exactly the plants the new rule was written to save. Nobody notices, because a pulled plant leaves no evidence of why it was pulled.

TAKES AS READ

a standing instruction can be replaced while work is still going on

VERDICT 151W

Three instructions run across the morning and each wants a different part of the nursery. Off-types out, then anything flowering early, then — once the rust is in the block — anything showing infection, which is a rule that keeps the clean plants standing rather than removing them. A plant can answer two of those at once, which is what makes the change cost something real. The plants scored are the ones either side of a change, because everything else in the row is wanted by neither rule and is left alone correctly by somebody who has not read anything. And the damage is invisible afterwards: a plant pulled to a withdrawn instruction leaves a gap in the row and no record of why, which is why the nursery's own rule is that the chalk is read at the start of every row rather than at the start of the morning.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

13.5 Six per cent, while the lorries arrive

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · CALLBACK · HOLD

SCENE 39W

The drying room has to bring this week's intake to six per cent moisture before any of it goes into the vault. Wet lots keep arriving off the field, and the dehumidifier is the only thing that answers them.

GUIDE 49W

Hold the room's moisture inside the band drawn on the dial. The band narrows as the week goes on, because seed that is nearly dry is spoiled by less. The dehumidifier is your control, and each lot that arrives is a load that keeps pushing until something answers it.

ASK 13W

Hold the drying room at six per cent while the week's intake arrives.

BEHIND THE BUTTON 146W

Why the band narrows. Early in the week the intake is wet and a point either way changes nothing. By the end the lots are close to store moisture, and the same point is the difference between seed that keeps for decades and seed that keeps for years.

Why a wet lot is not a step in the reading. A lorry does not raise the room by half a point and stop. It puts water into the air and goes on putting it there while it stands, so a room left alone does not settle — it keeps climbing.

Why this is the vault's real work. Cold and dry both slow the reactions that kill a seed, and they multiply. The freezer is easy: it holds itself. The moisture is the half that has to be held by somebody, every intake week, for the collection's life.

TAKES AS READ

seed is dried to a target moisture before it goes into store

VERDICT 142W

Every arrival here is a step in the rate rather than a step in the reading. A wet lot does not raise the room's moisture by a fixed amount; it adds water to the air and goes on adding it while it stands, so a room left alone climbs and keeps climbing. That is why the dehumidifier has to be moved to a new setting and left there, rather than nudged and released. The second half is the band narrowing. A tenth of a point of overshoot is nothing while the intake is at fourteen per cent, and it is the whole margin once the lots are near six. Both halves are the same lesson the vault is built on: cold and dry multiply, the cold holds itself, and the dry is held by whoever is on the drying floor that week.

TAKEAWAY 13W

A load that keeps arriving is not answered by a correction made once.

END OF THE DAY 13W

Three urgent things are three decisions, and the order is one of them.

What goes in the crossing block

PLAN CARD 97W

Thursday. The crossing block goes in tomorrow morning and it decides what the programme is working on until 2033 — a cross made in it is a line in 4 years and a release in 7. Volpe has her list, Qureshi has hers, and this year they overlap more than they ever have. The screen has 1 clean source of a second resistance gene, Fenn has 40 marked plants being threshed individually, and there is room in the block for 8 crosses. Today you fix the eight, and after tomorrow nothing in this season can be changed.

BRIEFING 11W

The block is drilled tomorrow. After that the parents are fixed.

WHAT YOU DECIDE 17W

Decide what the next seven years are made of, on the last day it can be changed.

14.1 Eight crossing commitments

HEREDITY & CROSSING · CROSSING HALL · A ROOM · ALLOCATE

SCENE 40W

The crossing board has eight programme slots for the next two years. Volpe wants elite yield crosses, Qureshi wants unrelated resistance material brought in, and the current elite line still needs the second resistance gene. Fenn's marked plants remain uncharacterised.

GUIDE 59W

Eight slots for two years, and more requests than slots. Watch the questions list rather than the crossing board: it shows what the programme can still answer under the current plan. Every package you commit is a question you can answer and one you have given up, and one of the requests can wait a season without costing anything.

ASK 16W

Which crossing packages keep the programme defended without spending the block on work that can wait?

BEHIND THE BUTTON 132W

What 'defended' means for a breeding programme. A single resistance gene is defeated eventually; a line carrying two is defeated much more slowly. Getting the second gene into the elite line is the work that keeps everything else the programme has built usable.

Why unrelated material is worth slots at all. Bringing in resistance from an unrelated source is slow and produces nothing marketable for years. It is also the only way to have a third gene when the second one fails, which is a decision about the decade rather than the season.

Why uncharacterised marked plants are a trap. They may be carrying exactly what is needed or nothing at all. Committing slots to them before characterisation is spending the block on a question that a cheap test could answer first.

TAKES AS READ

crosses consume greenhouse, generation and follow-up capacity for several seasons · closely related elite parents share more of the same genetic vulnerabilities

VERDICT 84W

The programme has two urgent genetic questions. One is to place the second resistance gene into the elite background farmers already use. The other is to put that gene into unrelated backgrounds so the programme does not rebuild the same single point of failure. Elite-to-elite yield crosses are useful but narrow the same base and consume a large package. Fenn's plants are promising but still uncharacterised. A finite crossing board therefore protects the two resistance questions first and deliberately leaves lower-priority packages for later.

TAKEAWAY 20W

A breeding plan is a finite set of genetic questions it chooses to answer, and every crossing commitment delays another.

14.2 How narrow the base has become

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A PERSON · TRACE

SCENE 39W

Ivanescu has the pedigree of six elite parents across the wall. Four share a recent grandparent, and all six carry the same resistance gene. The new rust race is already defeating that gene in commercial wheat east of Wellmere.

GUIDE 51W

Six parents, and five sources they could have drawn their resistance from. Name the source the pedigree actually shares, then keep only the parents whose resistance does not come through it. Six lines all reporting resistance is not six pieces of evidence if the resistance is one allele arriving six times.

ASK 15W

Which of these six parents give genuinely independent evidence that the programme has rust resistance?

BEHIND THE BUTTON 160W

Why agreement is not independence. Six trials of the same allele against the same rust race give one result, repeated. The programme's confidence came from the number of lines rather than from the number of sources, and those are different quantities whenever a pedigree has a shared ancestor in it.

What relatedness does to a breeding base. Four of the six share a grandparent, so a quarter of their genome is expected to be shared by descent alone. Crossing the best with the best concentrates exactly the alleles that made them the best, and narrows what is left to draw on when the pathogen changes.

Why the eastern outbreak is the whole point. A resistance gene is only useful against races that have not defeated it. The moment one race does, every line carrying that allele fails together — which is what a shared source means in practice, and why two independent sources are worth more than six correlated ones.

TAKES AS READ

related parents share more alleles than unrelated ones

VERDICT 128W

Four of the six parents descend from the 1998 line WM-311, and their resistance is that line's allele arriving four times over. Their agreement carries no more information than one of them does, which is why the programme's confidence was misplaced rather than merely optimistic: the number of lines went up and the number of sources did not. WM-604's resistance came in from a separate cross with a Chinese spring wheat, and the Anatolian introgression is a different gene at a different locus — two sources, and a race that defeats one need not defeat the other. This is what a narrowing base looks like from the inside. Everything in the nursery reports the same answer, and the reason is a common ancestor rather than a robust finding.

TAKEAWAY 18W

A programme that keeps crossing its best with its best is narrowing what it can still draw on.

14.3 DNA present, resistance working?

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A ROOM · CHOICE

SCENE 37W

Nakagawa can run the linked DNA marker at the two-leaf stage. On final candidates she can also measure resistance-gene mRNA after rust exposure and challenge leaves with the pathogen. The marker is four centimorgans from the gene.

GUIDE 55W

Four ways to order the evidence, and the question is which test belongs where. Ask of each what a marker can and cannot show. A linked marker can be separated from the gene by recombination. Even the gene's presence is not its expression: transcription, translation and regulation all sit between the sequence and the phenotype.

ASK 15W

Which evidence belongs early, and what should the final line still be required to show?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a linked marker can separate from a gene by recombination · transcription makes RNA from DNA and translation makes protein from mRNA · dominance, and why one allele is enough — taken as read

VERDICT 77W

The linked DNA marker is useful early because it can remove unlikely seedlings months before resistance can be tested. But recombination can separate that marker from the gene. Even direct DNA evidence only shows that a sequence is present. Transcription produces mRNA, translation produces protein, and regulatory interactions affect how much gene product appears and when. An RNA check after

rust exposure tests expression. The pathogen challenge then tests whether the whole pathway produces the resistant phenotype.

TAKEAWAY 21W

DNA can show what a plant carries; RNA and phenotype show whether that information is being used and whether it works.

14.4 What the trial can actually claim

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · CALLBACK · STRESS

SCENE 31W

Four seasons, one site, four replicates each. WM-712 is ahead by 11% in the 2 dry years and by 2% in the wet ones. The merchant's leaflet says "11% higher yielding".

GUIDE 74W

The claim on the leaflet rests on an assumption about how dry next season is, and the slider is that assumption — the soil-moisture deficit, where a low number is a wet year. Three candidate sentences are scored against two criteria. Take the slider down to a wet season and see which sentence is still true, because the one that reads best at the expected deficit is not the one that survives being wrong.

ASK 17W

Which sentence about WM-712 is still true if next season's rainfall lands anywhere in the plausible range?

BEHIND THE BUTTON 167W

What a trial measures. Four replicates at one site over four seasons separate a line from the pieces of ground it was grown on, and they do it under the weather those four seasons happened to bring. The number and its conditions are one object; quoting the number alone makes a different claim.

Why the expected season is the trap. At the four-season average deficit the bare eleven per cent needs the fewest words of qualification, which is exactly what a leaflet optimises for. Take the deficit down to a wet year and the sentence is wrong by a factor of five, and a buyer who paid for eleven and got two has been misled by a figure nobody fabricated.

What survives. A sentence that names the condition it was measured under stays true whatever next season does, and costs the merchant eleven more words on the leaflet. That is the trade being made, and genotype-by-environment interaction is the name for the reason it cannot be avoided.

TAKES AS READ

a trial measures lines under the conditions the trial happened in

VERDICT 142W

The eleven per cent is real and it belongs to two dry seasons. Set next season wet and the same line yields two per cent more than the standard, so a buyer who planned on eleven has been misled by a figure nobody fabricated. The conditional sentence gives up the headline and stays true across the whole rainfall range, which is what robustness means here: not the claim that scores best at the expected value, but the one that is still correct at the edges. The regional claim is worse than both — one site cannot support it at any rainfall, because nothing in the trial varied the region. Genotype-by-environment interaction is exactly this: a line's advantage is a property of the line and the conditions together, and a number quoted without its conditions is a different claim wearing the same digits.

TAKEAWAY 18W

A number from a trial carries the conditions it was measured under, or it is a different claim.

END OF THE DAY 18W

A crossing block is a decision about a decade, made in one morning with what is to hand.

What the season established

BEFORE THE DAY — TAG

Catch Nakagawa before the marker run starts

Nakagawa, the molecular laboratory lead, has the plate layout for this morning's marker run, and once the machine starts it is four hours and the layout is the only thing that says which well is which line. Catch her before she loads it, because a result with no layout is ninety-six numbers and no names.

PLAN CARD 99W

Friday, and the season is set. The crossing block went in yesterday, WM-712 was released with its weakness written on the label, and the drifted accessions are on a list that will take 15 years to work through. Fenn's marked plants are drying in 40 paper bags with his handwriting on them. The board meets on Monday and Qureshi has to say what happened here in terms somebody who funds a gene bank will understand. Today you settle what the season established, what it cost to find out, and which of the things done under pressure are worth keeping.

BRIEFING 16W

The block is in. The rust is here. The collection is smaller than the catalogue says.

WHAT YOU DECIDE 11W

Say what Wellmere knows now, what it cost, and what changes.

15.1 What the catalogue says and what is there

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · BALLPARK

SCENE 37W

90 accessions were genotyped. 30 had drifted, and every 1 of those had been regenerated at least twice. The other 60, regenerated once or not at all, still matched. About 9,000 accessions have 2 or more regenerations.

GUIDE 68W

Turn the sample into a number for the group it came from, and only for that group. Every drifted accession in the sample had been regenerated twice or more, and the collection holds about nine thousand of those. Two of the tiles belong to the whole collection rather than to the high-risk group, and putting one of them in gives the number the board must not be told.

ASK 11W

How many accessions does this season's sample actually put at risk?

BEHIND THE BUTTON 132W

What the sample supports. All thirty drifted accessions came from the twice-regenerated group, and none of the sixty from outside it drifted. That identifies a risk factor cleanly. What it does not do is measure the drift rate of accessions nobody sampled.

Why the whole-collection figure is the wrong one. Forty-one thousand accessions times any rate estimated from this sample would be a claim about material that was never tested, and most of the collection has never been regenerated at all — the group where nothing drifted.

What the honest sentence looks like. A rate, the group it applies to, and the number in that group: about a third of roughly nine thousand twice-regenerated accessions, which is around three thousand and is not the same statement as a third of the collection.

TAKES AS READ

a sample of a collection supports an estimate about the collection

VERDICT 133W

Thirty of the ninety sampled accessions had drifted, and every one of the thirty came from the twice-regenerated group. Within the sample that group is 30 of 90, which is not the rate to use — the rate that matters is drifted-over-sampled inside the high-risk group itself, and applied to the roughly 9,000 accessions the collection holds in that group it gives about 3,000. The 41,000-accession figure is a different claim entirely, about material that was never tested and most of which has never been regenerated, which is the group where nothing drifted at all. Both numbers are arithmetic on the same sample; only one of them is a statement the evidence supports. The board will quote whichever sentence it is given, which is why the group has to travel with the fraction.

TAKEAWAY 17W

A sample can identify a high-risk group without proving the exact fraction affected in the entire collection.

15.2 One site, one year, again

FIELD TRIALS & SELECTION · FIELD LABORATORY · A PERSON · CHOICE

SCENE 44W

The trial is scored and the selected lines are chosen. The summary sheet now goes to a board that will remember one headline number. This was a dry season at one site, and the selected plots looked much better than the full replicated mean.

GUIDE 55W

Four ways to write one result. Ask of each where the conditions sit relative to the number. A footnote is not the same sentence. The selected-plot mean is worse as a headline, because those plots were chosen partly for their ground. And withholding the result treats a conditional finding as useless rather than as conditional.

ASK 8W

How should this season's trial result be summarised?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a trial's conclusions belong to the conditions it ran under

VERDICT 83W

A yield number becomes misleading when the conditions that produced it are stripped away. This season's gain belongs to one site and a dry year. The selected-plot mean is even less defensible as a headline because those plots were chosen partly for favourable ground and weather. Putting site and season in the same sentence as the gain is not cosmetic caution; it defines the claim being made. A later season can test generality, but it does not make the current conditional result useless.

TAKEAWAY 14W

The sentence that survives being quoted out of context is the one worth writing.

15.3 11 years, and what they are for

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM · CLOUD

SCENE 37W

Toft's canopy, flowering and weather records now span 12 seasons. Nobody has ever asked for the whole series. This year two of those records explained why the highest-yielding plots separated from the rest during a dry finish.

GUIDE 67W

The cloud is the twelve seasons of summer rainfall, one point each, and the band that counts as an ordinary season is drawn across it. Place the middle bar and the two sigma bars to report where the seasons sit and how widely they scatter, then use what the archive offers to sharpen the estimate. A single year has no position until the distribution behind it exists.

ASK 22W

Report where an ordinary summer sits and how widely the seasons scatter, and say whether this year is inside the ordinary band.

BEHIND THE BUTTON 155W

Why one season cannot be read alone. This summer delivered 118 millimetres. On its own that number means nothing at all; against twelve seasons it is the second driest, which is what turns a yield difference into evidence about dry-finish performance.

What the two bars are for. The middle bar is the station's estimate of an ordinary season and the outer pair is the uncertainty on it, one sigma either side. A pair placed honestly holds roughly two thirds of the points between them, which is what one sigma means and is worth checking on the plot rather than taking on trust.

Why a series cannot be started late. Extending it backwards is possible only where somebody else kept a record; the airfield's rain gauge can be borrowed and the canopy states of past seasons cannot, because the crops are gone. That asymmetry is the whole argument for measuring routinely in years when nobody asks.

TAKES AS READ

a long record allows a season to be compared against others · resource allocation: defence, growth and yield out of one budget — taken as read

VERDICT 168W

A season is a number until there is a distribution behind it, and then it becomes a position. Twelve summers of rainfall at this site scatter around roughly 196 millimetres with a one-sigma width near 34, so this year's 118 sits more than two sigma low — the second driest in the series and comfortably outside the ordinary band. That is what licenses the sentence Toft wants to write: WM-712's advantage appeared in a season that was unusual in a stated, measured way. The station's own record is short, so borrowing the airfield's longer gauge series narrows the estimate of the middle without moving it much, and correcting the exposure difference

between the two gauges shifts it without narrowing anything. Neither operation invents information about this summer; both change how confidently the station can say what an ordinary summer is. And the canopy half of the record cannot be recovered at all, which is why a routine measurement in a year nobody asks about is the asset here.

TAKEAWAY 15W

A measurement series is worth more each year, and it can only be started early.

15.4 What came with the gene

HEREDITY & CROSSING · CROSSING HALL · A ROOM · CALLBACK · BALLPARK

SCENE 41W

Every backcross line carrying the resistance allele is 12 to 15 centimetres shorter than its susceptible siblings. The donor parent was a short landrace, while the recurrent elite parent is tall. Volpe wants to know what travelled with the target gene.

GUIDE 52W

The scoring sheet counts how often the marker and the height gene come apart. Work out that fraction from the plants that recombined, and read it as a distance along the chromosome. Two tiles belong to the whole population and two to the recombinant class, and only one pairing is the frequency.

ASK 13W

How far apart are the resistance gene and the height locus, in centimorgans?

BEHIND THE BUTTON 156W

Why a distance is the useful form. A recombination frequency of one per cent between two loci is defined as one centimorgan of map distance, so the fraction measured in a population converts straight into how far apart the two genes sit. That is what tells a breeder whether the drag can be broken at all.

Why a chromosome travels in stretches. Crossing over happens at a limited number of points per chromosome per meiosis, so a wanted allele arrives with a block of its neighbours. The closer a neighbour sits, the more generations it takes to separate — and the shortness here is a neighbour, not a cost of resistance.

What the number is for. Six recombinants in two hundred is three centimorgans, which is close enough that finding a line carrying resistance without the height block means screening hundreds of plants rather than dozens. Volpe's answer is therefore a screening budget, not a shrug.

TAKES AS READ

genes sit at fixed places on chromosomes · dominance, and why one allele is enough — taken as read

VERDICT 152W

The short plants are not paying a price for being resistant. They are carrying a block of the donor's chromosome that came across with the resistance allele, and a height gene is inside that block. How often the two come apart is measurable: 6 of the 200 backcross plants carry one without the other, so the recombination frequency is $6 \div 200 = 0.03$, and 1 per cent recombination is defined as 1 centimorgan of map distance. Three centimorgans is close linkage. It means the drag can be broken — the two loci do separate — but only by screening enough plants to catch a 3 per cent event, which is why linkage drag is a budget question rather than a permanent feature of the line. The alternative explanations are all testable and all fail: resistance

costing growth would shorten every resistant plant in every background, and backcrossing does not reduce vigour.

TAKEAWAY 14W

A chromosome is inherited in stretches, so a wanted gene arrives with its neighbours.

END OF THE DAY 19W

A collection is only as good as the last time each accession was grown, and now Wellmere knows which.

Card text only — no options, boards or answers. 13,590 words of card prose across 61 stops. Generated from themes/seedbank by tools/make-cardbook.mjs.